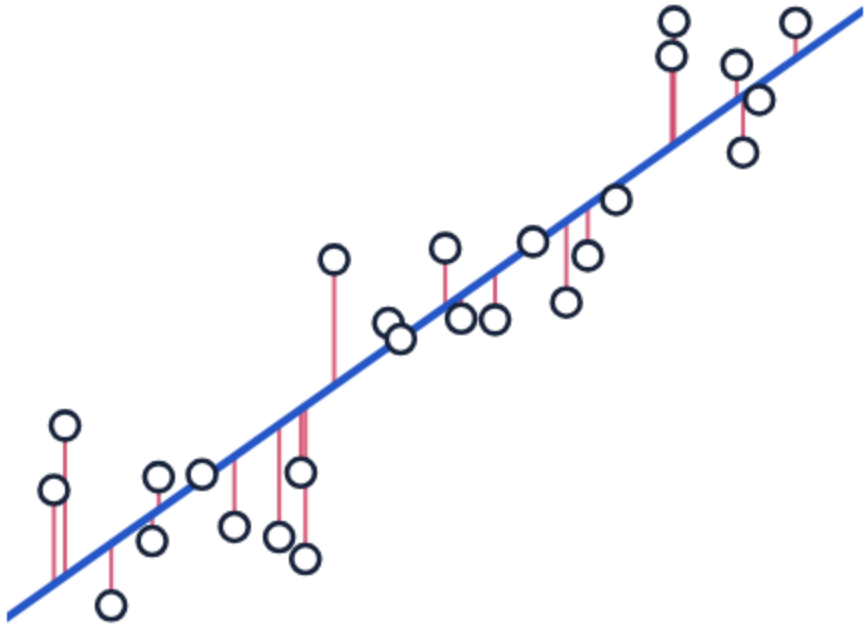
Lecture 1 / 180 minutes

Linear regression from first principles



Press → or space to advance

0–25 min

0

From human perception to machine learning

What are we trying to learn?

Code to be human



Can you recognise who this is?

Can you write an algorithm that does the same?

Human perception



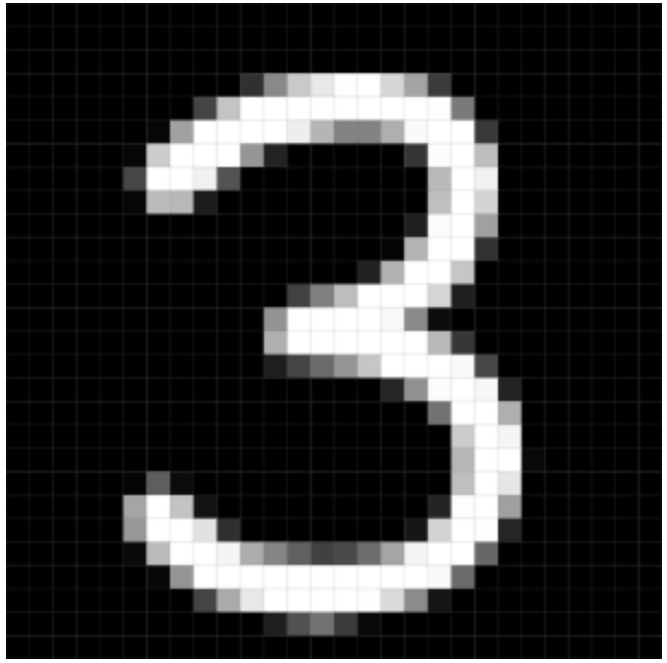
Faces, voices, objects, language.

What information do we use to make these decisions?

Algorithm vs human

Let’s try something simpler.

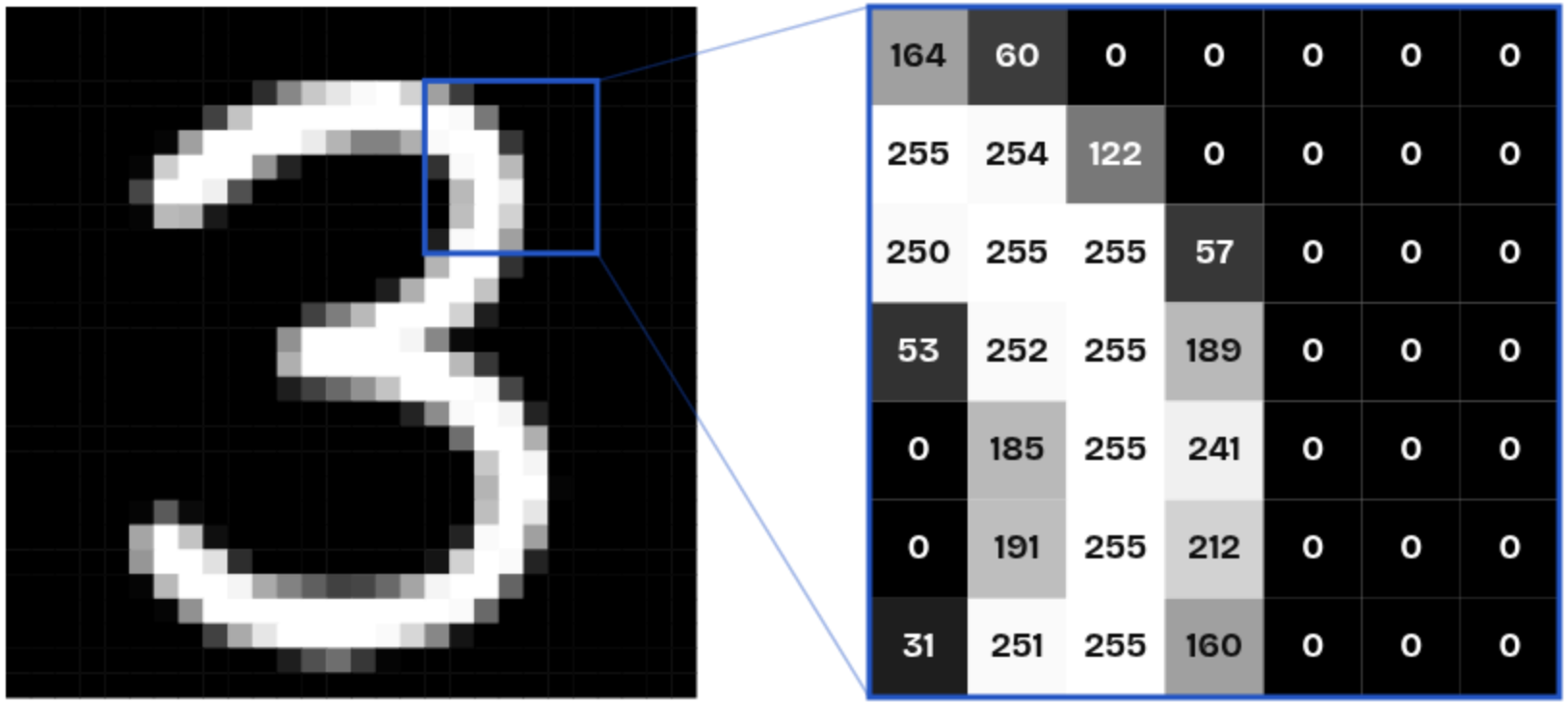




MNIST: Modified National Institute of Standards and Technology dataset

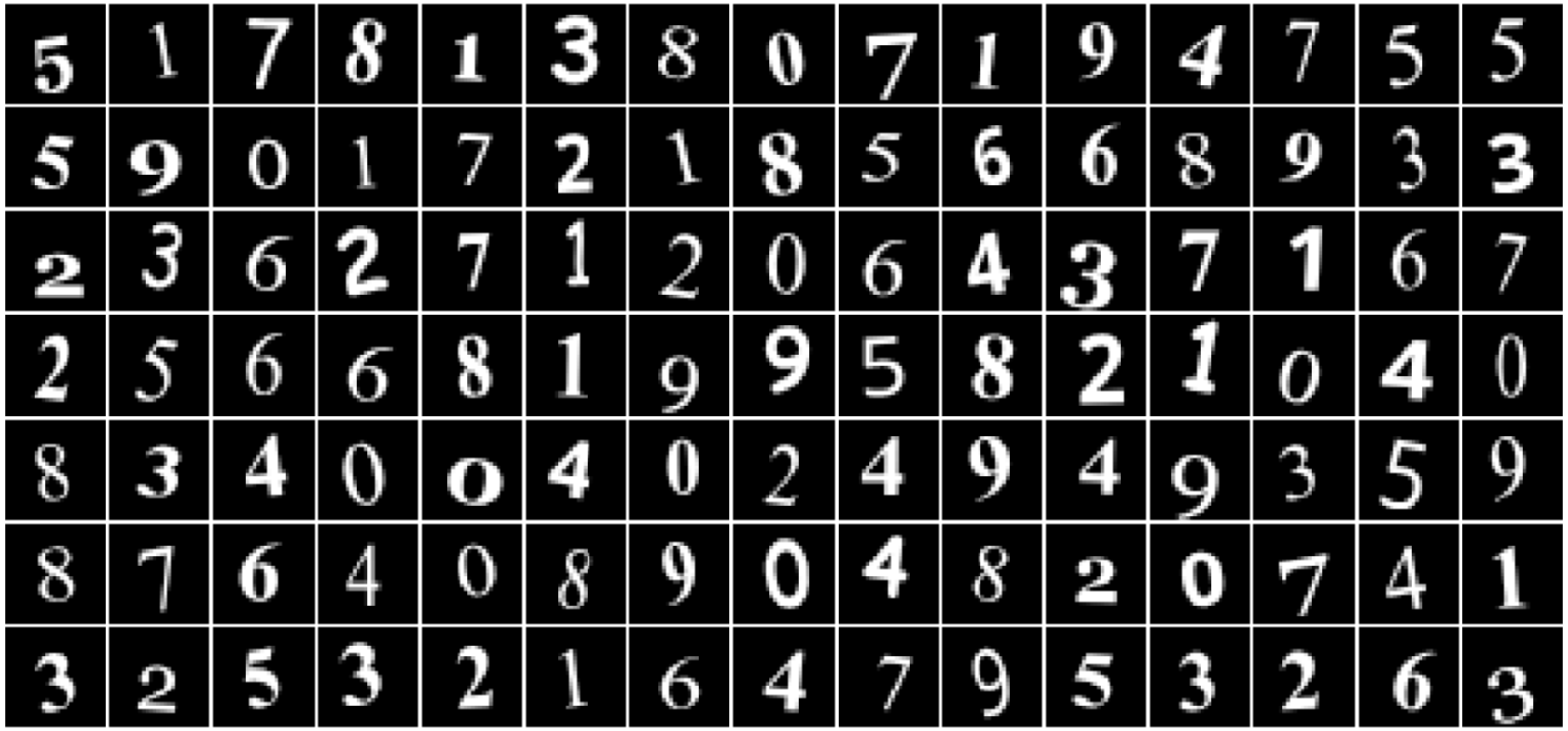
What does a computer see?

An image is a numerical array. $x \in \{0, \ldots, 255\}^{H \times W}$



For an MNIST digit: $H = W = 28$, so 784 numbers.

Machine learning: handwritten digits



Can examples help us learn to recognise new digits?

Machine learning: natural images



Can a model learn object categories from labelled images?

Machine learning: text

From: prize-desk@w1nner.biz

You have been selected!!!

Dear user, CONGRATULATIONS, you won a free phone. Click now to claim your $1000 reward.

spam

From: r.mehta@university.edu

Lecture 2 materials

Hi all, the slides and the exercise sheet for Thursday are on the course page. See you in room 204.

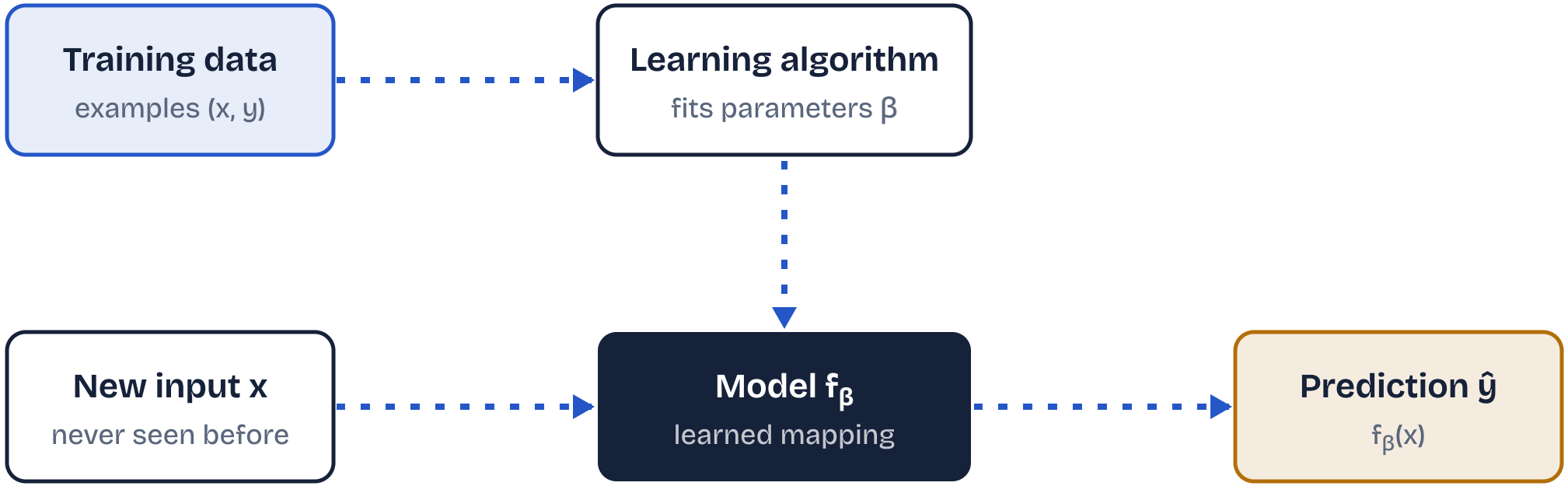
not spam

Can a model distinguish spam from legitimate email?

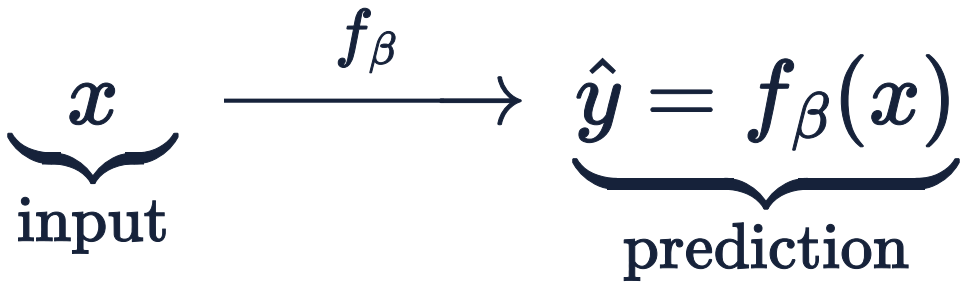
Different data modalities

Modality	Numerical representation
Image	Pixel values
Audio	Signal samples
Text	Token representations
Tabular	Feature vectors

The machine learning approach



A mathematical black box



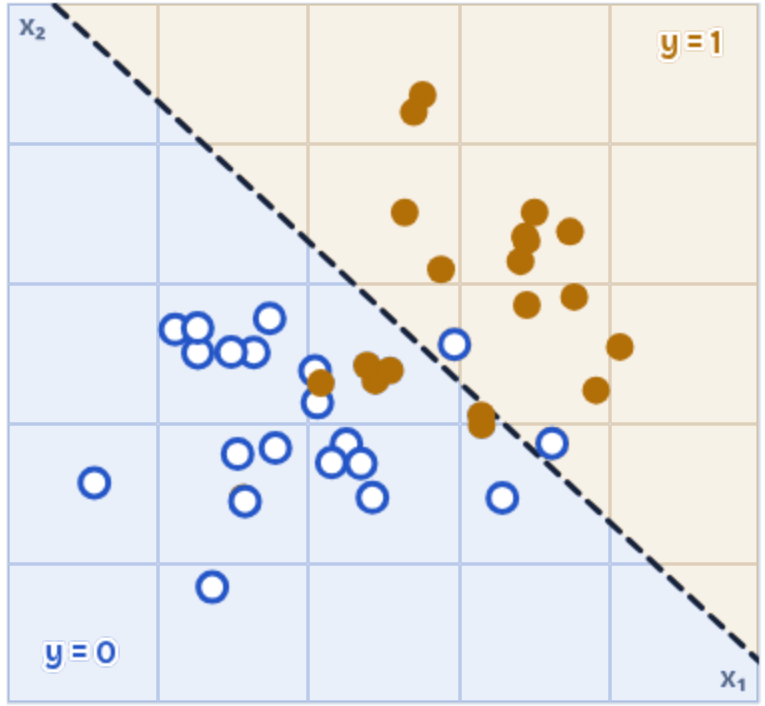
The training data determine the model parameters β.

Supervised learning

Inputs *and* targets:

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{n}$$

Learn f_β to predict y from x.

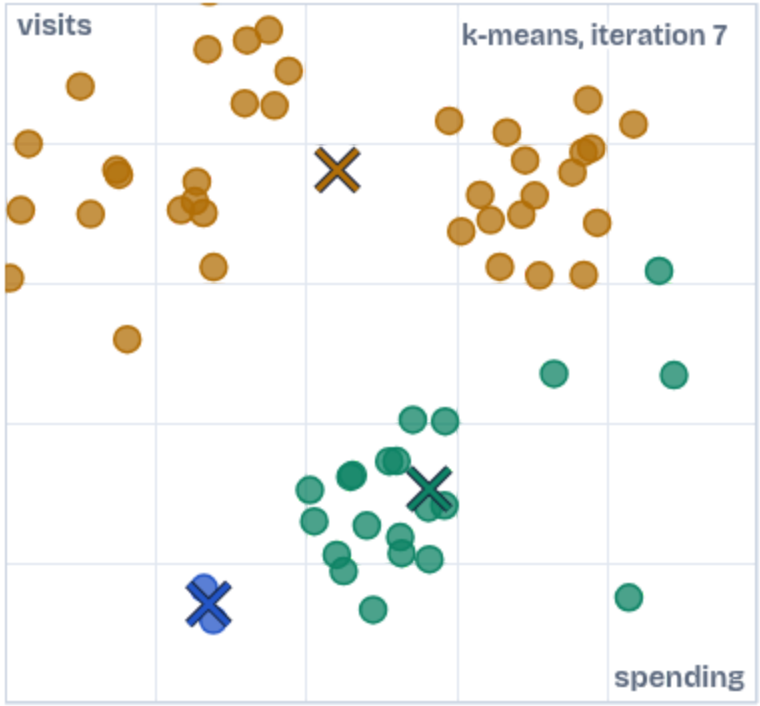


Unsupervised learning

Inputs, no target labels:

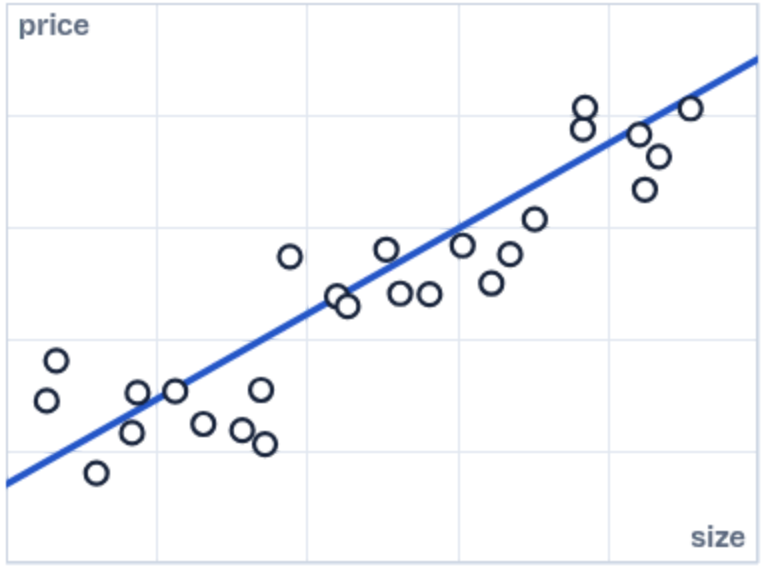
$$\mathcal{D} = \{x_i\}_{i=1}^{n}$$

Example: group customers with similar behaviour.



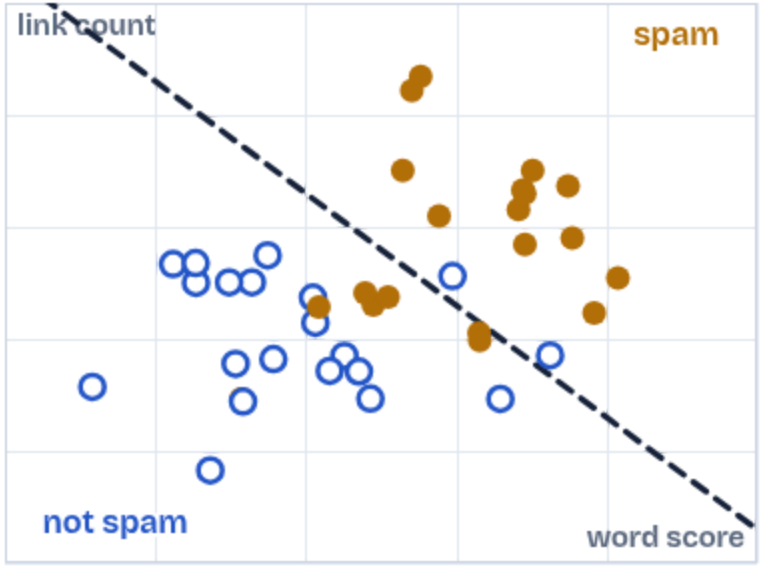
Two supervised learning tasks

Regression



Predict a number: house price

Classification



Predict a class: spam / not spam

Banking



You are hired by a bank to **automate loan pricing.**

Bank case study: past data

Customer	Income (₹ L)	Score	Loan (₹ L)	Rate (%)
A	8	690	4	12.5
B	12	720	6	11.2
C	5	650	3	14.1
D	15	780	8	9.8
E	10	710	5	11.5

Income and loan amount in ₹ lakhs.

Bank case study: your task

A new customer applies for a loan.

$$x = (\text{income},\ \text{credit score},\ \text{loan amount})$$

Predict an interest rate: $\hat{y} \in \mathbb{R}$.

Bank case study: loan default

Customer	Income (₹ L)	Score	Loan (₹ L)	Default
A	8	690	4	No
B	12	720	6	No
C	5	650	3	**Yes**
D	15	780	8	No
E	10	710	5	No
F	4	620	3	**Yes**

Same features, different tasks

Loan pricing

$$y \in \mathbb{R}$$

regression

Loan default

$$y \in \{0, 1\}$$

classification

Regression or classification?

1. Predict the price of a used car. regression
2. Predict whether an email is spam. classification
3. Predict tomorrow's number of mall visitors. regression (a count)

A more difficult question

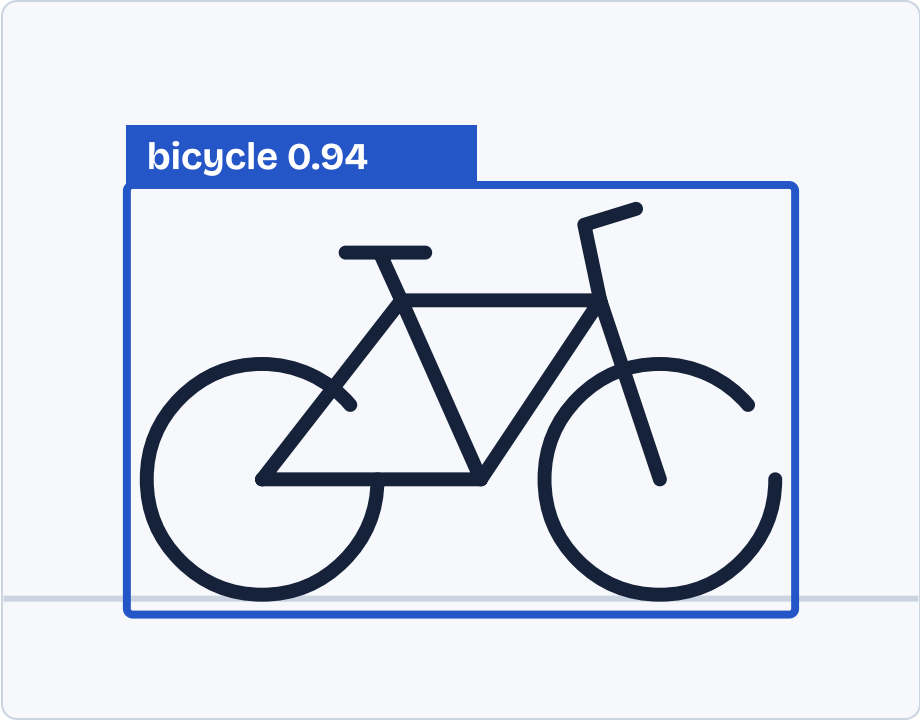
1. Predict a movie's rank in a recommendation list. ranking
2. Predict the next word in a sentence. token prediction
3. Find where a pedestrian appears in an image. localisation

These have more structure than a scalar or a binary label.

Object detection

Goal: identify objects and locate them.

Example: **bicycle**



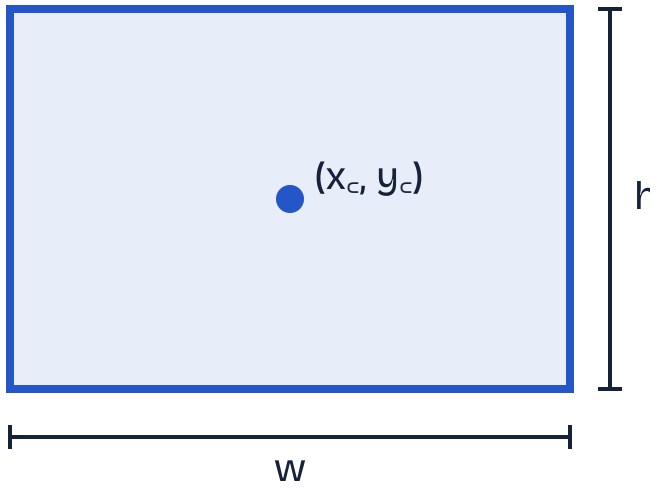
Object detection: two outputs

Object class

$$c \in \{1, \ldots, K\}$$

Bounding box

$$b = (x_c, y_c, w, h) \in \mathbb{R}^4$$



Object detection combines tasks

$$\underbrace{\text{object category}}_{\text{classification}} + \underbrace{\text{bounding-box coordinates}}_{\text{regression}}$$

Supervised learning: formal setup

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{n}, \qquad x_i \in \mathbb{R}^d, \quad y_i \in \mathcal{Y}$$

$$f_\beta : \mathbb{R}^d \longrightarrow \mathcal{Y}$$

Goal: low prediction error on **new** observations.

Regression vs classification

Regression	Classification
$\mathcal{Y} \subseteq \mathbb{R}$	$\mathcal{Y} = \{1, \ldots, K\}$
Numerical target	Categorical target
Interest rate	Default / no default

Two ways to understand a model

Geometric	Probabilistic
$y \approx X\beta$	$Y = X\beta + \varepsilon$
projection, rank	likelihood, uncertainty

Both lead to linear regression.

25–72 min

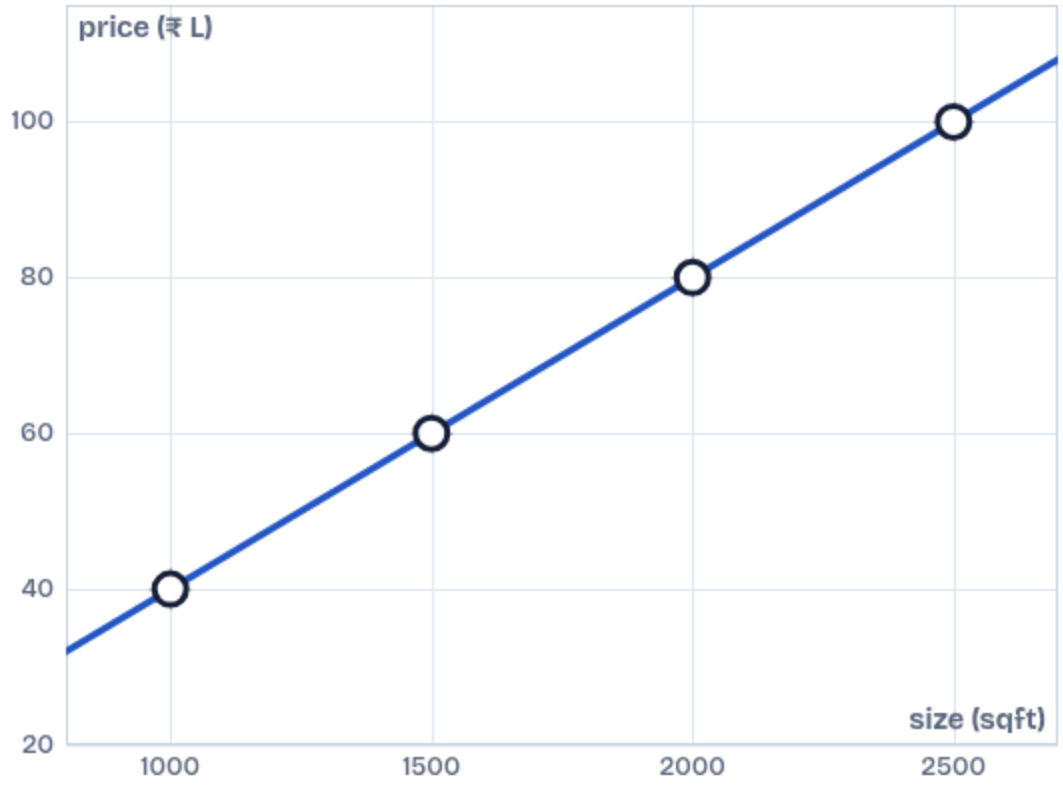
1

Linear regression

From solving equations to fitting data

Motivating example: house prices

Size (sqft)	Price (₹ L)
1000	40
1500	60
2000	80
2500	100

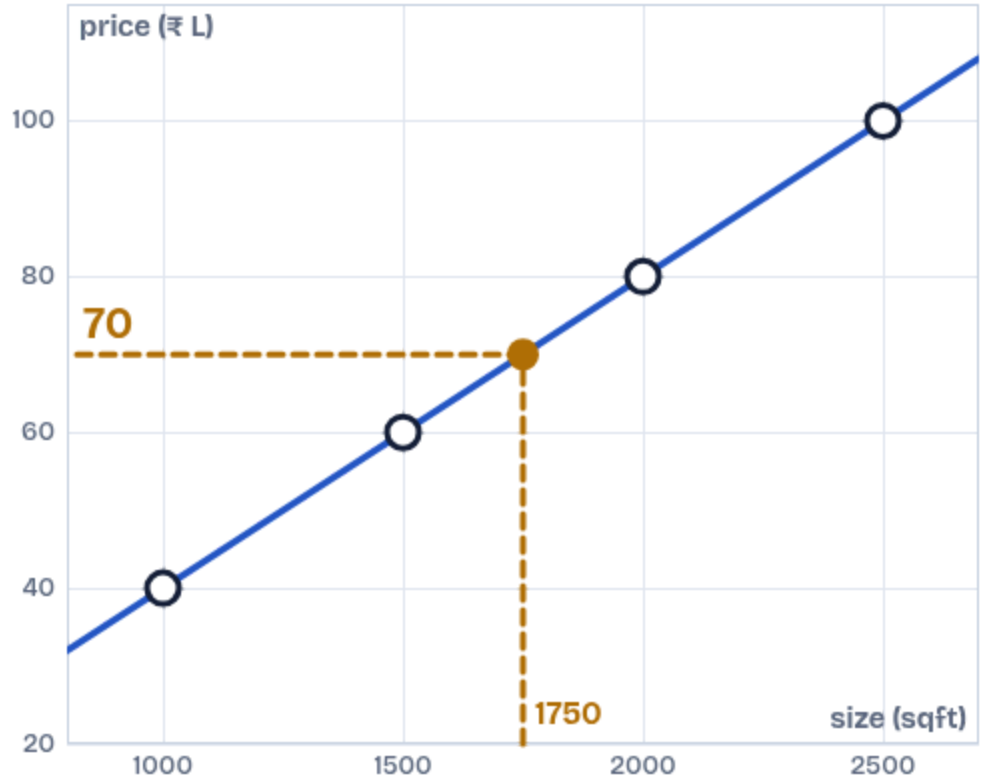


Predict a new house price

How much for a **1750 sqft** house?

$$\hat{y} = 0.04 \times 1750 = \boxed{70 \text{ lakhs}}$$

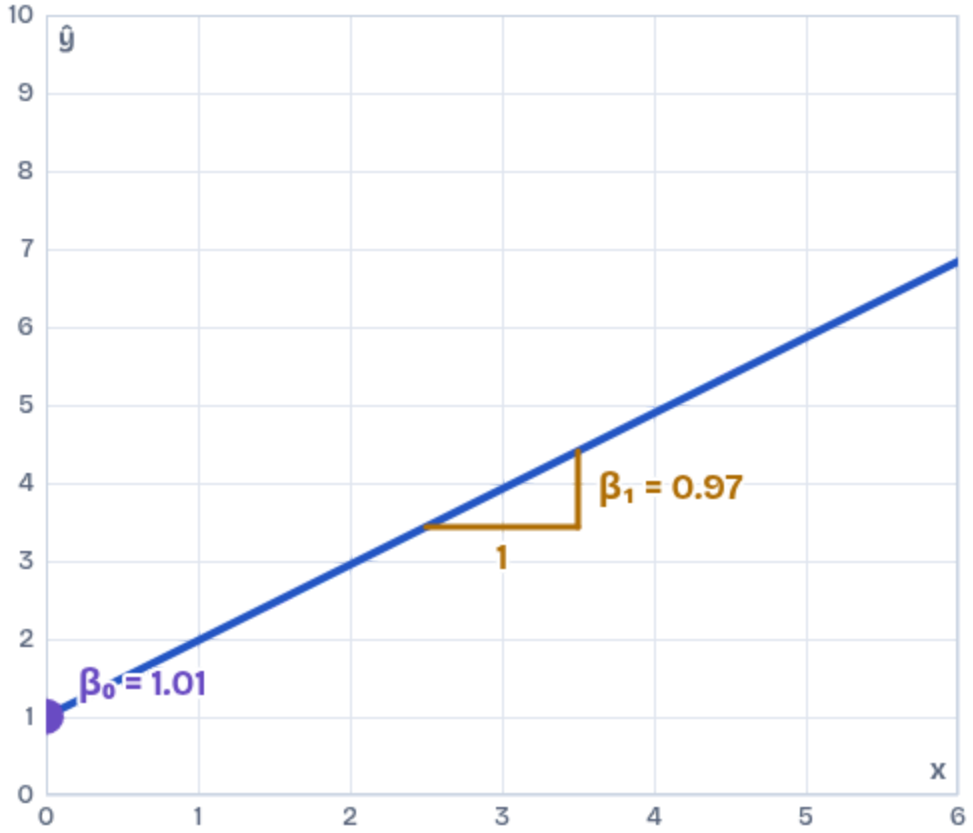
We implicitly assumed a linear relationship.



The equation of a line

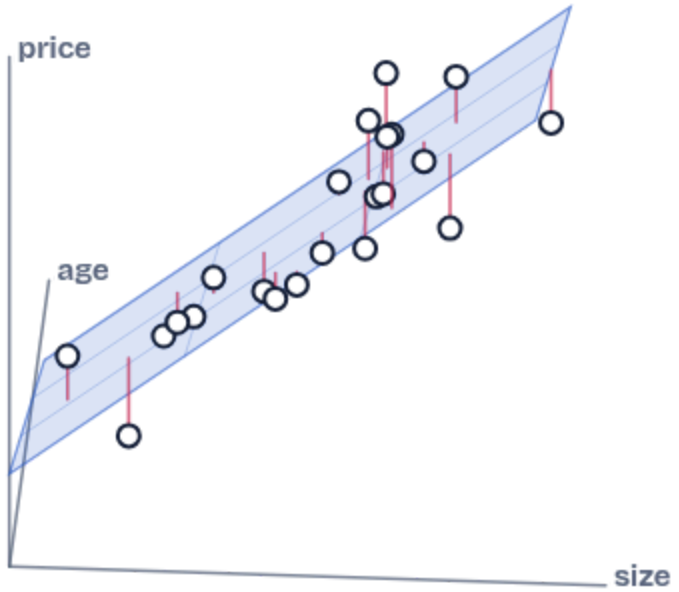
$$\hat{y} = \beta_0 + \beta_1 x$$

β_0	intercept
β_1	slope
x	input feature
$\hat{y}$	predicted response



What if we have multiple features?

$$\hat{y} = \beta_0 + \beta_1 x_1 + \cdots + \beta_d x_d$$



Two features: the line becomes a plane. Size, age → price.

Linear regression in matrix form

$$\underbrace{\begin{bmatrix} 1 & x_{11} & \cdots & x_{1d} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \cdots & x_{nd} \end{bmatrix}}_{X \in \mathbb{R}^{n \times (d+1)}} \underbrace{\begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_d \end{bmatrix}}_{\beta} \approx \underbrace{\begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}}_{y}$$

The intercept as a feature

For $\boldsymbol{x}_i \in \mathbb{R}^d$, define

$$\tilde{\boldsymbol{x}}_i = \begin{bmatrix} 1 \\ \boldsymbol{x}_i \end{bmatrix} \in \mathbb{R}^{d+1}$$

Row i of $\boldsymbol{X}$ is $\tilde{\boldsymbol{x}}_i^\top$.

Exercise 1: find the parameters

$$\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

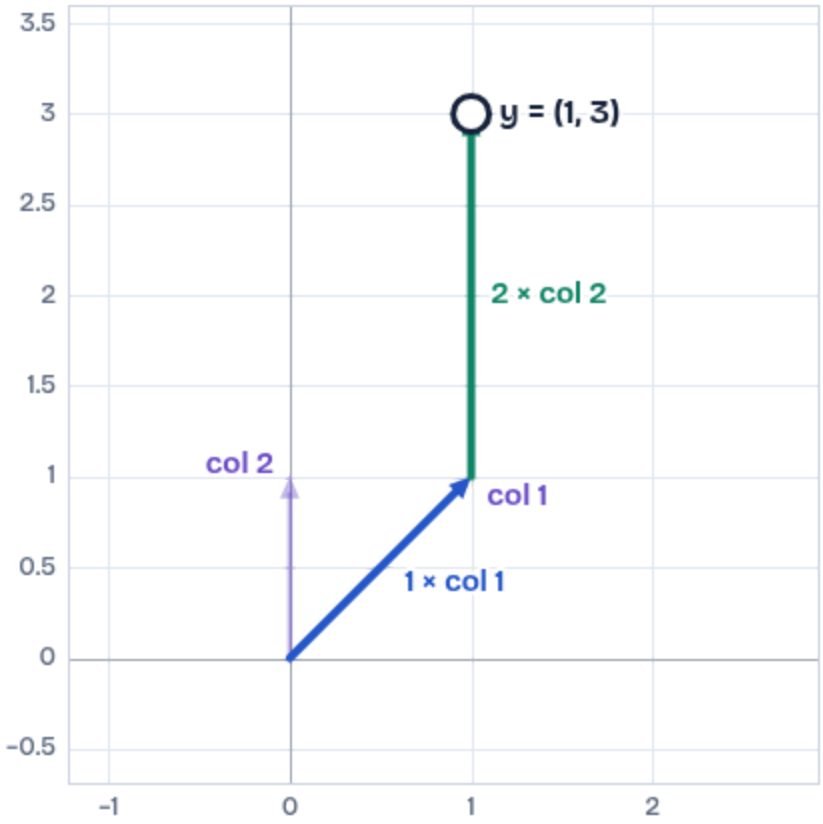
Find β_0 and β_1.

3 minutes

Exercise 1: solution

$$\beta_0 = 1, \qquad \beta_0 + \beta_1 = 3$$

$$\boxed{\beta_0 = 1, \quad \beta_1 = 2}$$



Column picture: 1 × column 1 + 2 × column 2 lands exactly on y.

When is the solution unique?

For a square system $X\beta = y$:

$$\det(X) \neq 0 \implies \beta = X^{-1}y$$

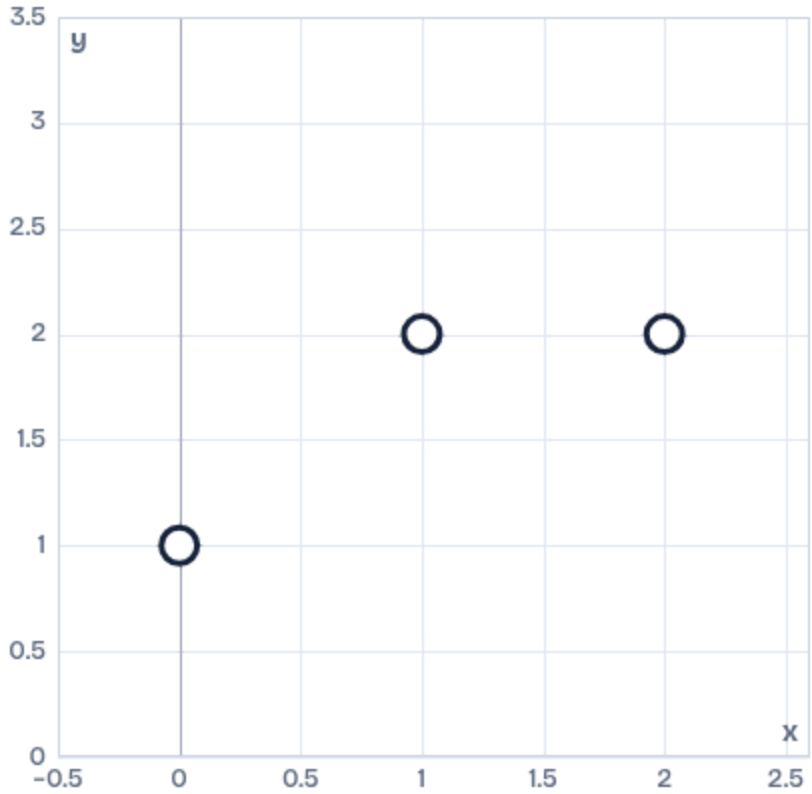
What if there are more observations than parameters?

Exercise 2: more equations than unknowns

$$\begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

Can all three equations hold exactly?

5 minutes



Why an exact solution does not exist

The first two equations imply

$$\beta_0 = 1, \qquad \beta_1 = 1$$

The third would then require

$$1 + 2(1) = 2 \quad \text{✗}$$

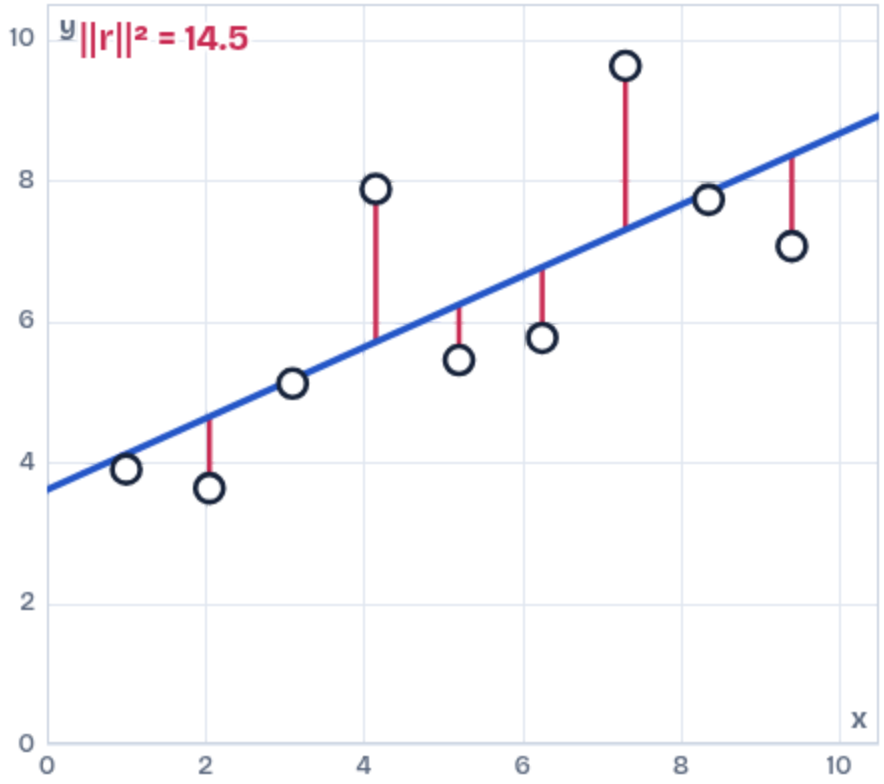
Instead, minimise the error

Residual vector

$$r(\beta) = y - X\beta$$

Ordinary least squares

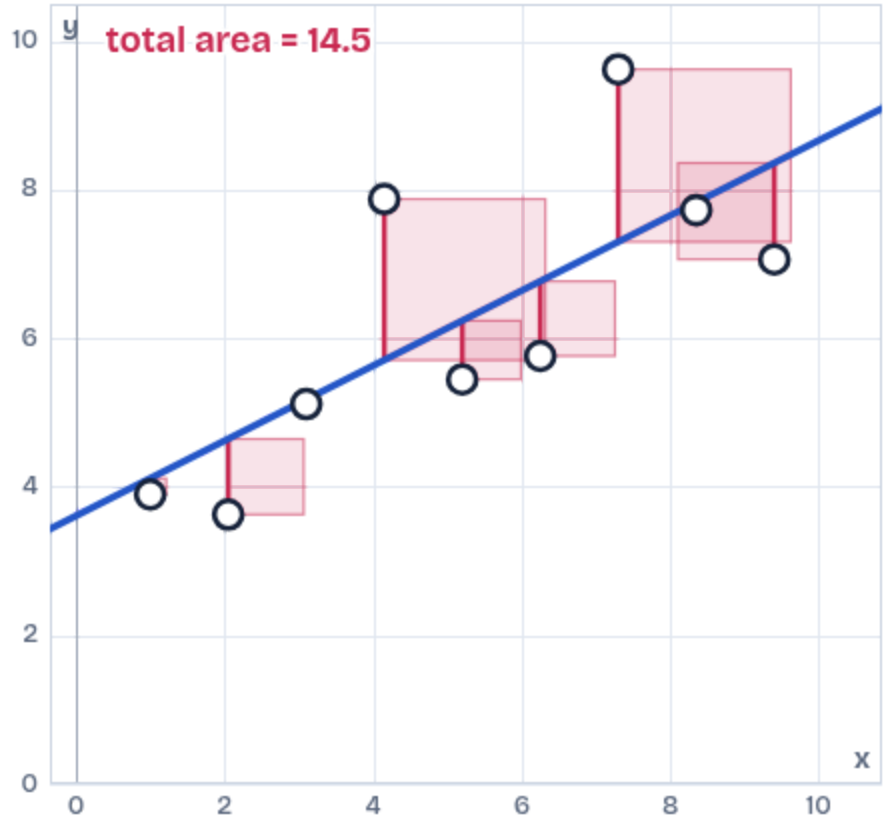
$$\hat{\beta} \in \arg\min_{\beta} \|y - X\beta\|_2^2$$



Ordinary least squares

$$\|y - X\beta\|_2^2 = \sum_{i=1}^{n} (y_i - \tilde{x}_i^\top \beta)^2$$

The total area of the squares.

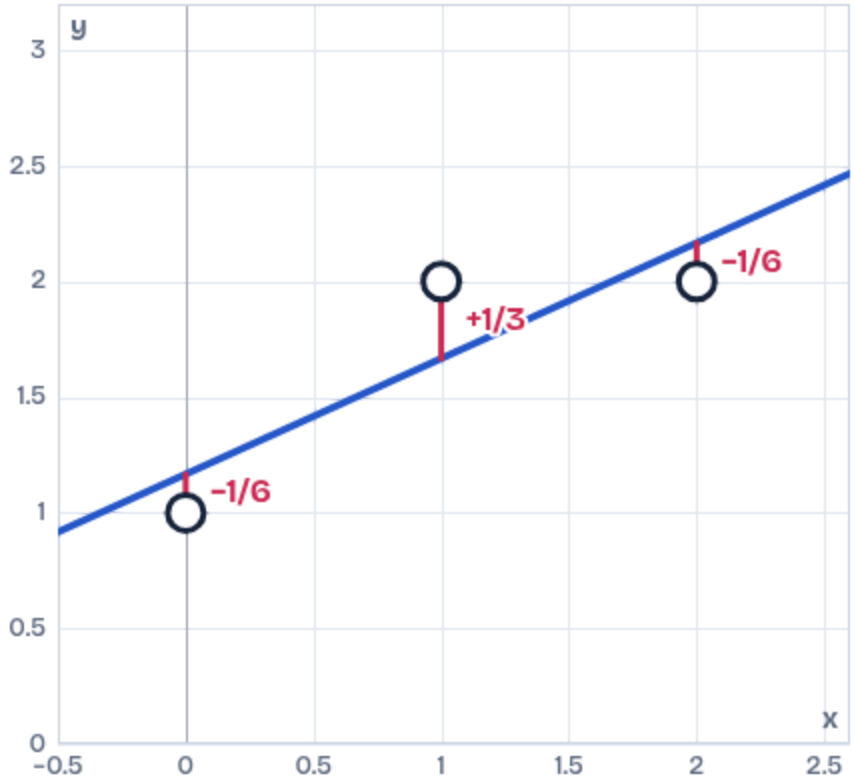


Exercise 2: best-fitting line

$$(x_i, y_i) \in \{(0,1), (1,2), (2,2)\}$$

Find the line with minimum squared error.

$$\boxed{\hat{y} = \frac{7}{6} + \frac{1}{2}x}$$



Deriving the OLS solution

$$L(\beta) = (y - X\beta)^\top (y - X\beta)$$

$$\nabla_\beta L(\beta) = -2X^\top (y - X\beta)$$

The normal equations

$$X^\top (y - X\hat{\beta}) = 0$$

$$\boxed{X^\top X\hat{\beta} = X^\top y}$$

The closed-form OLS estimator

$$\text{If } \mathbf{rank}(X) = d + 1,$$

$$\boxed{\hat{\beta} = (X^\top X)^{-1} X^\top y}$$

Why do we need full column rank?

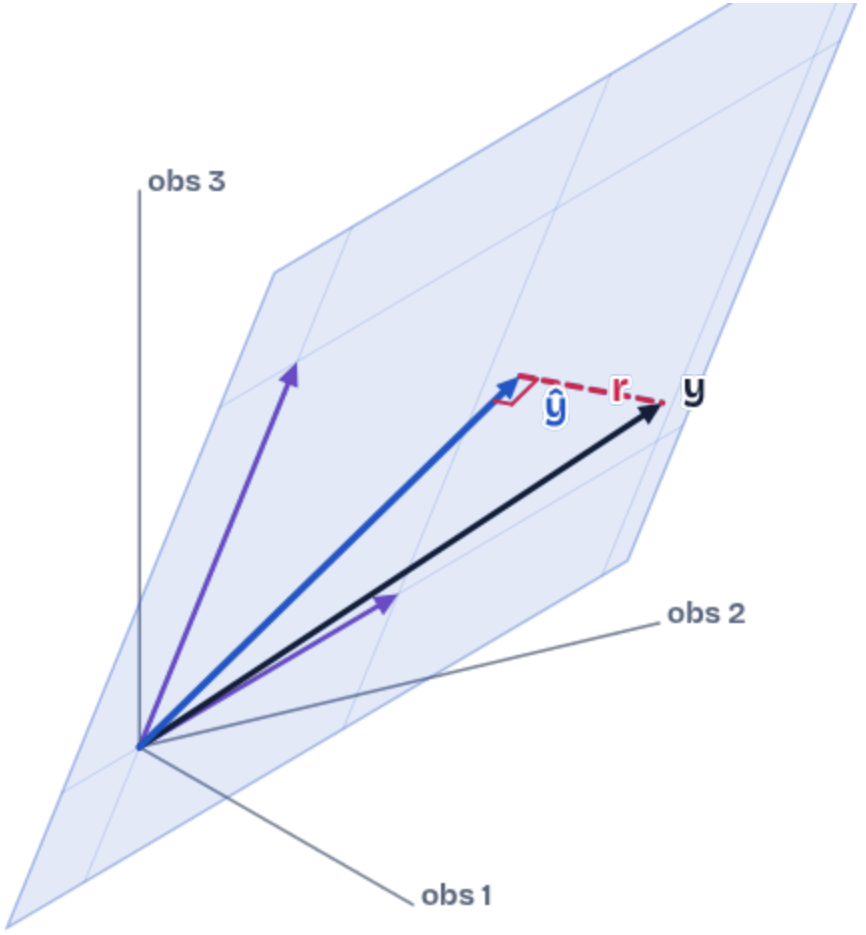
The geometry of least squares

$$\hat{y} = X\hat{\beta} \in \text{col}(X)$$

$$\boxed{\hat{r} = y - \hat{y} \perp \text{col}(X)}$$

$\hat{y}$ is the orthogonal projection of y.

Drag to rotate



The projection matrix

$$\hat{y} = P_X y, \qquad \boxed{P_X = X(X^\top X)^{-1} X^\top}$$

$$P_X^2 = P_X, \qquad P_X^\top = P_X$$

Projecting twice changes nothing.

A geometric check

$$X^\top \hat{r} = 0$$

Are residuals orthogonal to every feature column?

Yes, including the intercept column: $\sum_i \hat{r}_i = 0$.

72–97 min

2

Multicollinearity and identifiability

When the coefficients stop being unique

What if $X^\top X$ is singular?

$$\det(X^\top X) = 0$$

The inverse formula cannot be used.

Does this mean predictions are impossible?

Exercise 3: dependent features

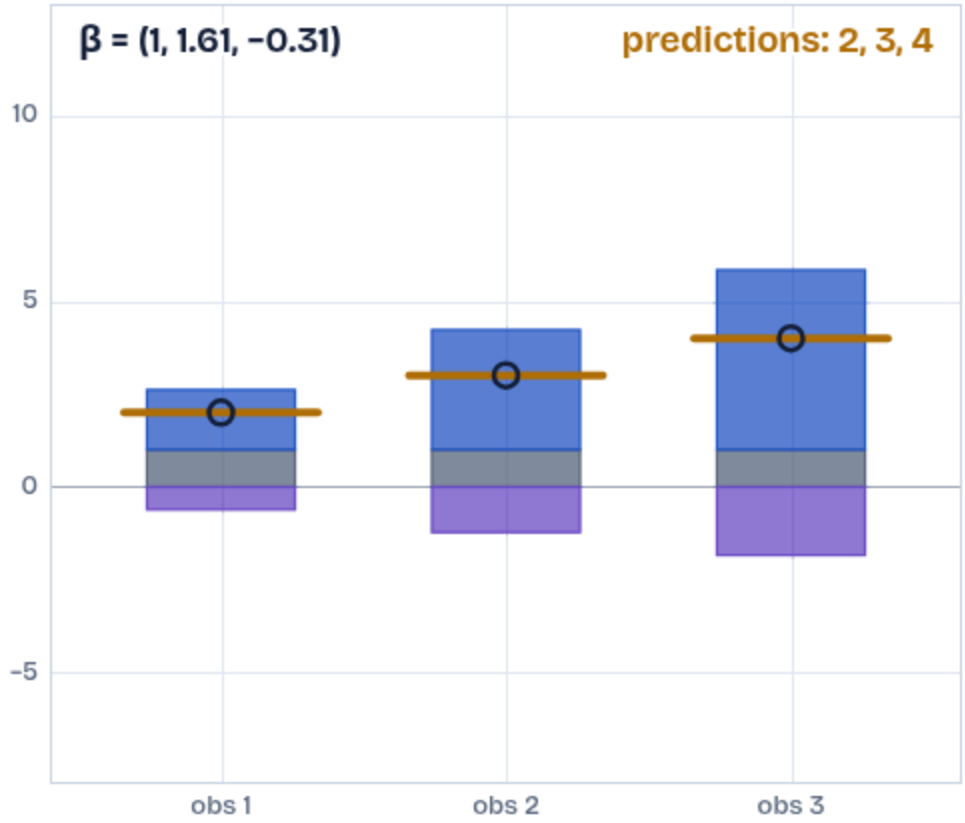
$$X = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 4 \\ 1 & 3 & 6 \end{bmatrix}, \qquad y = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$$

Column 3 is twice column 2.

Exercise 3: two different solutions

$$\beta^{(1)} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \qquad \beta^{(2)} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$\boxed{X\beta^{(1)} = X\beta^{(2)} = y}$$



Bars: β_0 (grey), $\beta_1 x_1$ (blue), $\beta_2 x_2$ (violet). Amber: prediction.

Identifiability

β is identifiable from $X\beta$ if

$$X\beta = X\beta' \implies \beta = \beta'$$

This holds precisely when $\mathbf{null}(X) = \{0\}$.

Why predictions can survive

If $v \in \mathbf{null}(X)$, then

$$X(\beta + v) = X\beta$$

Different coefficients, identical fitted values.

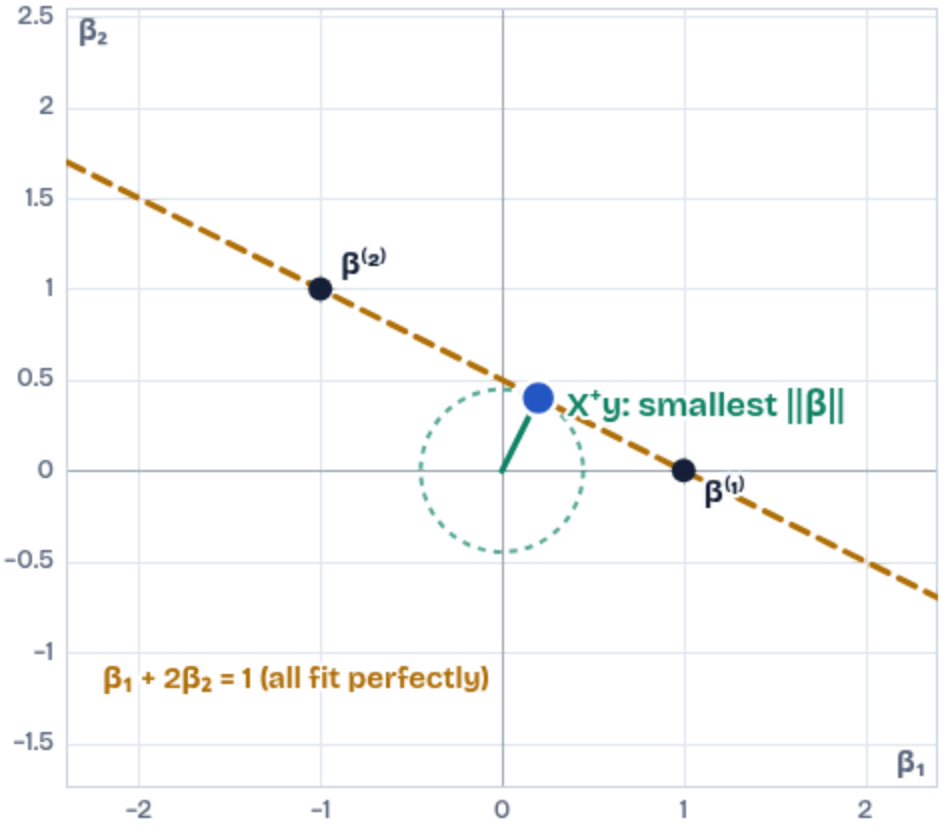
Here $v = (0, -2, 1)$.

How do we fit a singular system?

$$\boxed{\hat{\beta} = X^{+} y}$$

Moore–Penrose pseudoinverse

The **minimum-norm** least-squares solution.



107–127 min

3

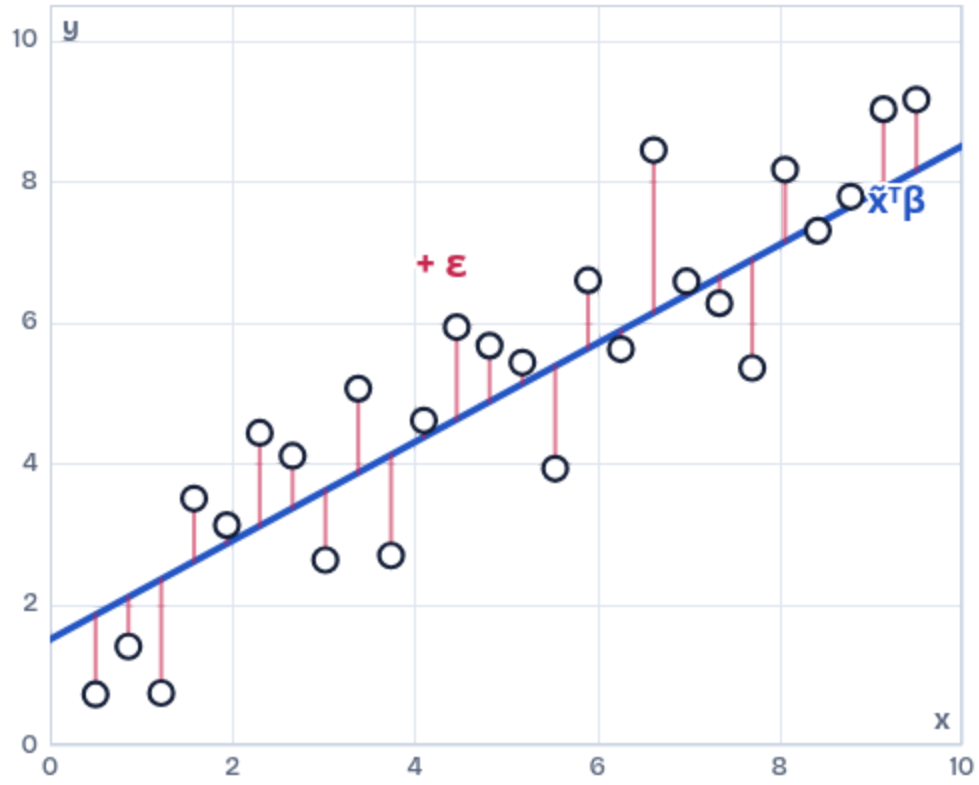
The probabilistic view

Maximum likelihood and coefficient uncertainty

A second view of linear regression

$$y_i = \tilde{x}_i^\top \beta + \varepsilon_i$$

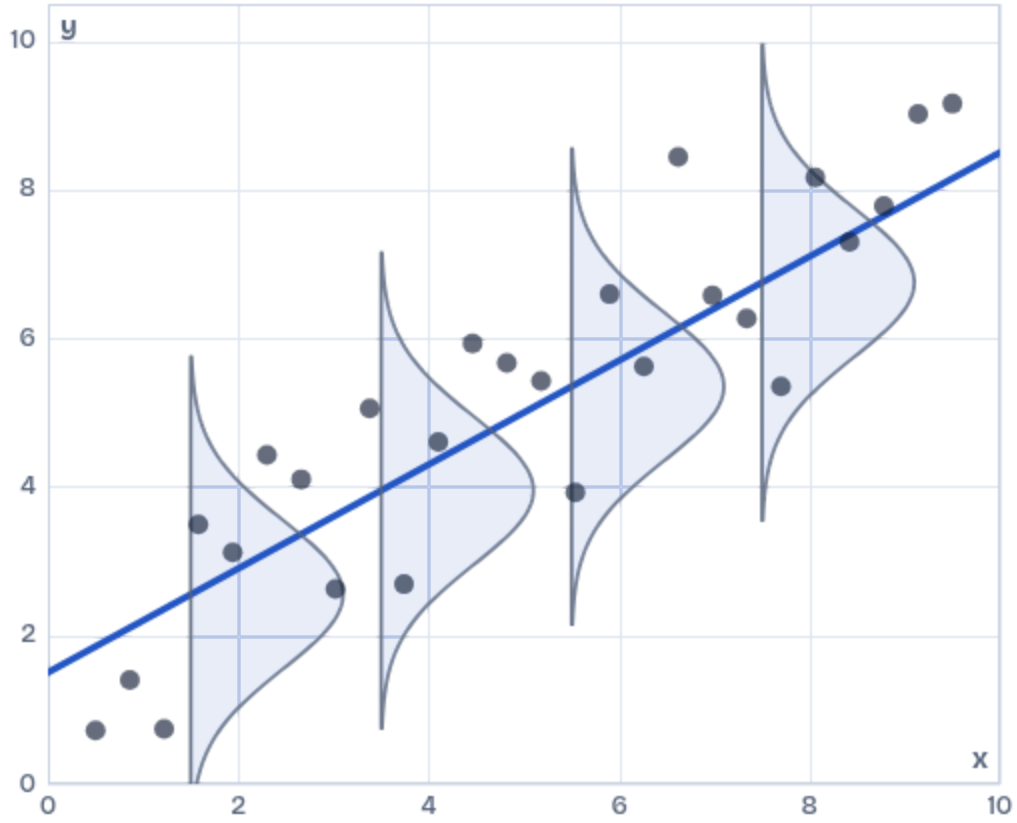
What should we assume about ε_i?



Gaussian noise model

$$\varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$$

$$Y_i \mid x_i \sim \mathcal{N}(\tilde{x}_i^\top \beta, \sigma^2)$$



Likelihood of one observation

$$p(y_i \mid x_i, \beta, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \tilde{x}_i^\top \beta)^2}{2\sigma^2}\right)$$



Large prediction errors are less likely.

Likelihood of the dataset

By conditional independence,

$$L(\beta, \sigma^2) = \prod_{i=1}^{n} p(y_i \mid x_i, \beta, \sigma^2)$$

Why is taking a logarithm helpful?

Products become sums; tiny numbers stop underflowing.

Log-likelihood

$$\log L = \underbrace{-\frac{n}{2}\log(2\pi\sigma^2)}_{\text{no } \beta} \underbrace{-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \tilde{x}_i^\top \beta)^2}_{\text{squared error}}$$

Which term depends on β?

Exercise 4: MLE vs OLS

For fixed $\sigma^2 > 0$:

$$\arg\max_{\beta} \log L = \arg\min_{\beta} \sum_{i=1}^{n} (y_i - \tilde{x}_i^\top \beta)^2$$

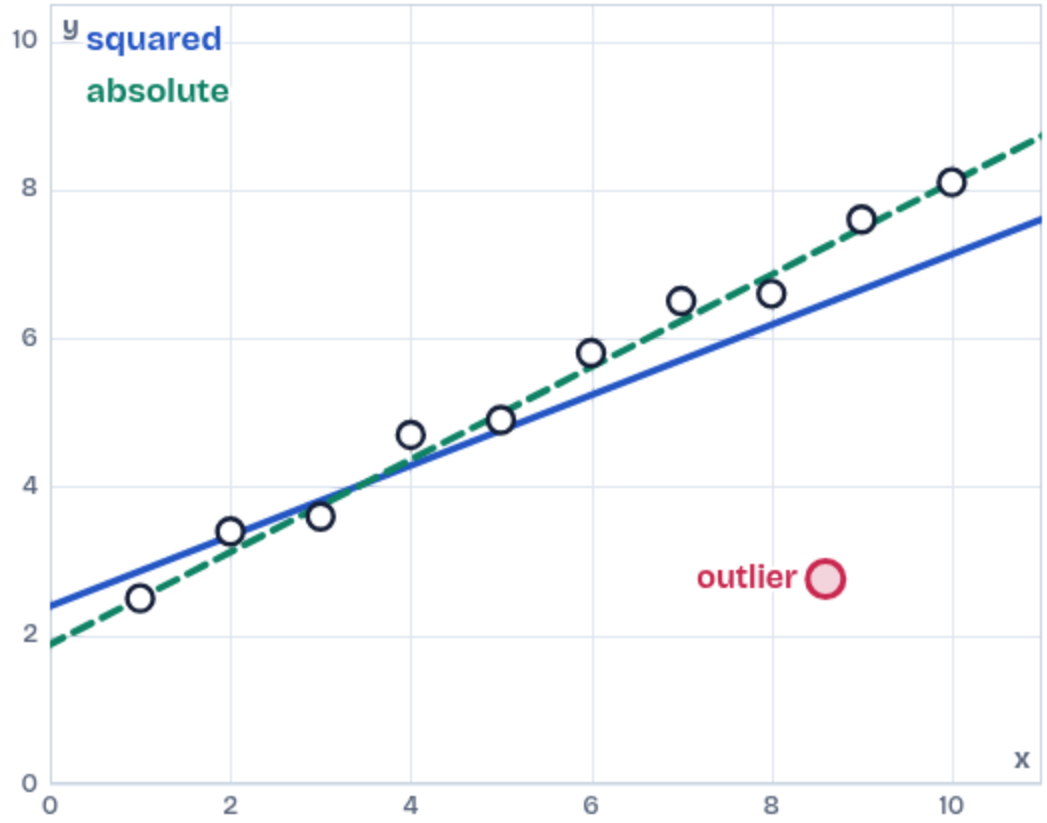
$$\boxed{\hat{\beta}_{\text{MLE}} = \hat{\beta}_{\text{OLS}}}$$

Does squared loss always apply?

Gaussian errors → squared error

Laplace errors → absolute error

The noise assumption determines the loss.

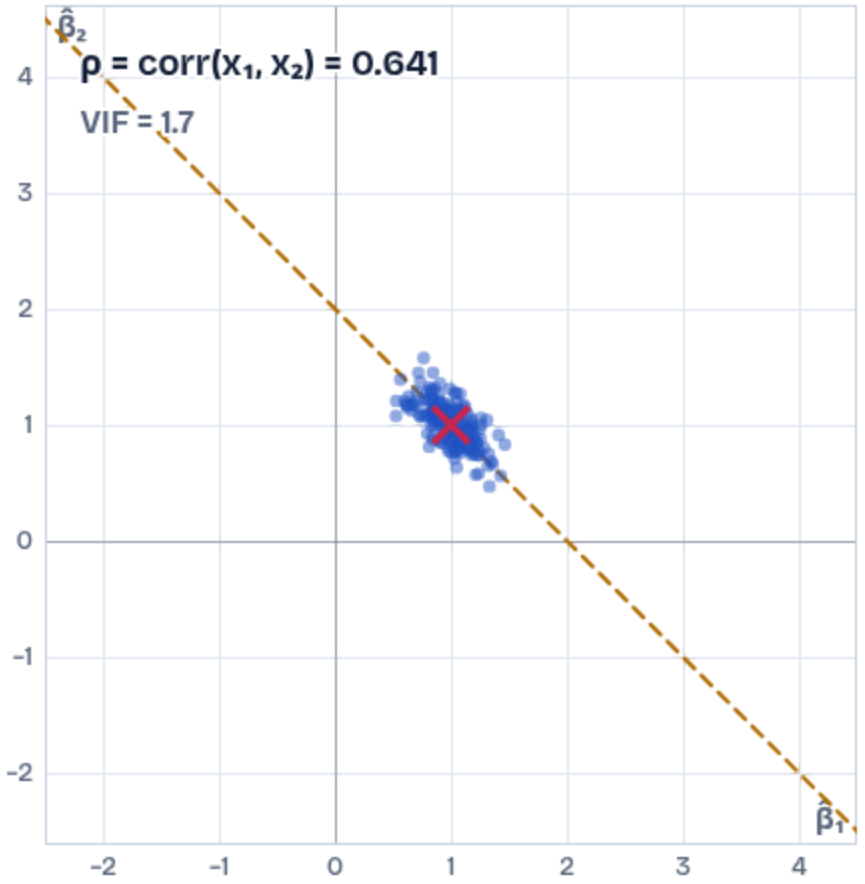


Near multicollinearity

Suppose $x_2 \approx 2x_1$.

$X^\top X$ invertible but ill-conditioned.

Small changes in data, large changes in coefficients.



200 datasets, true β = (1, 1). Each dot is one dataset's estimate.

What is R_j^2 in the VIF?

Original model:

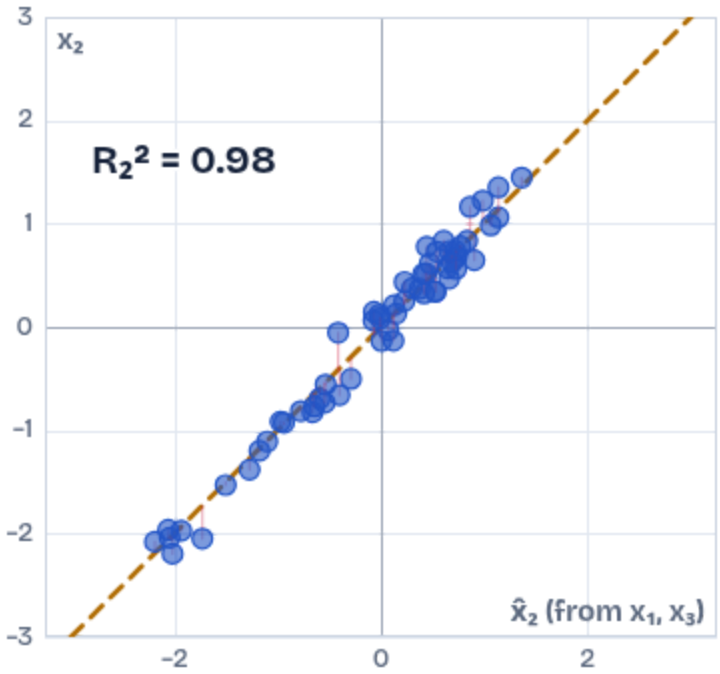
$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \varepsilon_i$$

How much of x_2 can the other features explain?

Fit an **auxiliary regression**:

$$x_{i2} = \alpha_0 + \alpha_1 x_{i1} + \alpha_3 x_{i3} + u_i$$

$$R_2^2 = 1 - \frac{\sum_i (x_{i2} - \hat{x}_{i2})^2}{\sum_i (x_{i2} - \bar{x}_2)^2}$$



- $R_2^2 = 0$: other features explain none of the variation in x_2.
- $R_2^2 = 0.96$: they explain 96% of it.
- $R_2^2 = 1$: x_2 is perfectly explained by the others.

We are predicting x_2, not the original target y!

Variance inflation factor (VIF)

$$\boxed{\mathrm{VIF}_j = \frac{1}{1 - R_j^2}}$$

Why it matters. Under the classical homoscedastic model,

$$\mathrm{Var}(\hat{\beta}_j \mid X) = \underbrace{\frac{\sigma^2}{\sum_{i=1}^{n}(x_{ij} - \bar{x}_j)^2}}_{\text{variance without collinearity}} \cdot \underbrace{\frac{1}{1 - R_j^2}}_{\mathrm{VIF}_j}$$

$$R_2^2 = 0.96 \;\Rightarrow\; \mathrm{VIF}_2 = \frac{1}{1 - 0.96} = \boxed{25}$$

- variance of $\hat{\beta}_2$ inflated × 25
- standard error inflated × $\sqrt{25} = 5$

What if $R_2^2 = 1$? $\mathrm{VIF}_2 = \infty \implies$ **β_2 is not uniquely identifiable.**

Variance of OLS coefficients

If $\mathbf{Var}(\varepsilon \mid X) = \sigma^2 I$ and X has full column rank,

$$\boxed{\mathbf{Var}(\hat{\beta} \mid X) = \sigma^2 (X^\top X)^{-1}}$$

Near-collinearity makes some diagonal entries large.

Variance inflation factor

Regress x_j on the other features; get R_j^2.

$$\mathrm{VIF}_j = \frac{1}{1 - R_j^2}$$

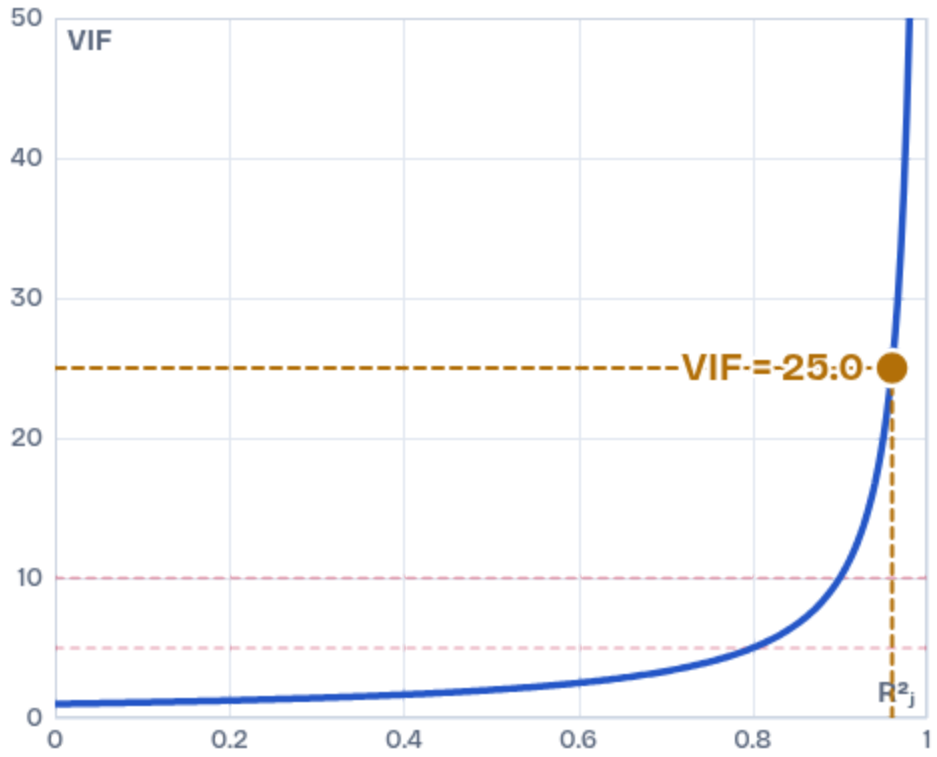


Exercise 5: calculate the VIF

Suppose $R_j^2 = 0.96$.

$$\text{VIF}_j = \frac{1}{1 - 0.96} = \boxed{25}$$

Variance × 25, standard error × 5.



127–180 min



Can we trust our regression model?

Standard errors, t-tests, confidence intervals and diagnostics

A question about the coefficients

Suppose linear regression estimates

$$\boxed{\hat{\beta}_1 = 2.4}$$

Does this tell us that the true coefficient is 2.4?

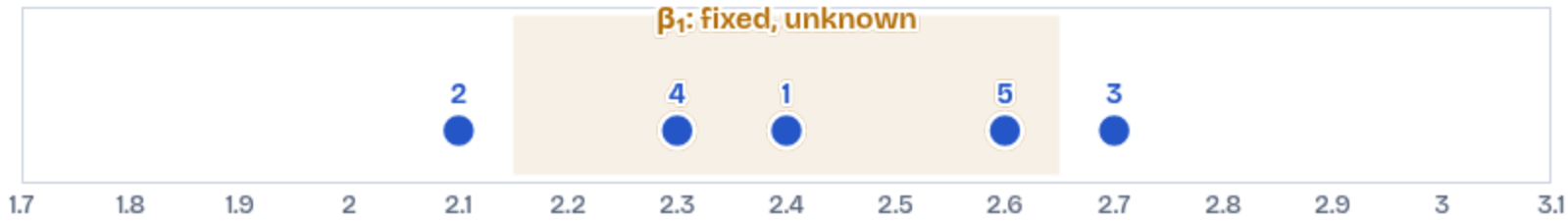
No. We estimated β_1 from a **finite, noisy sample**.

What would happen if we collected a different sample?

Different samples, different estimates

Repeat the same study with new observations:

Sample	1	2	3	4	5
$\hat{\beta}_1$	2.4	2.1	2.7	2.3	2.6



The estimate changes from sample to sample.

$\hat{\beta}_1$ is a random variable; β_1 is the fixed, unknown parameter.

Numbers are illustrative.

The statistical model

$$\boxed{y = X\beta + \varepsilon}$$

$X \in \mathbb{R}^{n \times p}$ has full column rank.

$$\mathbb{E}[\varepsilon \mid X] = 0, \qquad \mathbf{Var}(\varepsilon \mid X) = \sigma^2 I_n$$

- $\beta \in \mathbb{R}^p$: true, unknown coefficients
- ε: random errors with common variance σ^2

Why is the OLS estimate random?

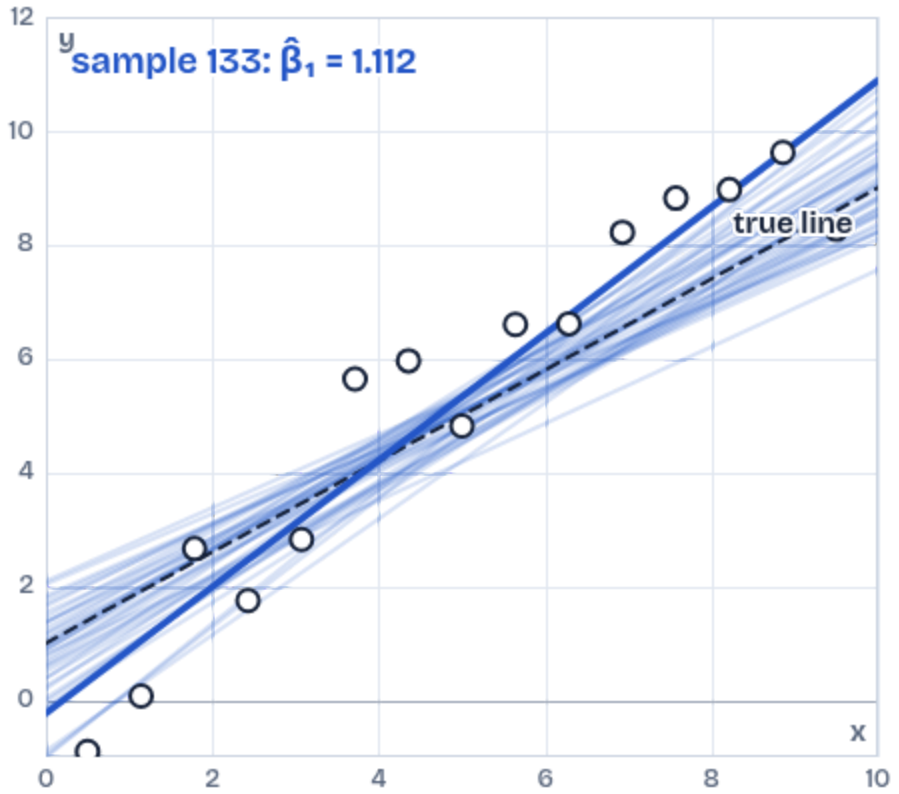
$$\hat{\beta} = (X^\top X)^{-1} X^\top y$$

Substitute $y = X\beta + \varepsilon$:

$$\hat{\beta} = \beta + (X^\top X)^{-1} X^\top \varepsilon$$

$$\boxed{\hat{\beta} - \beta = (X^\top X)^{-1} X^\top \varepsilon}$$

Random errors produce random estimates.



Mean and variance of the OLS estimator

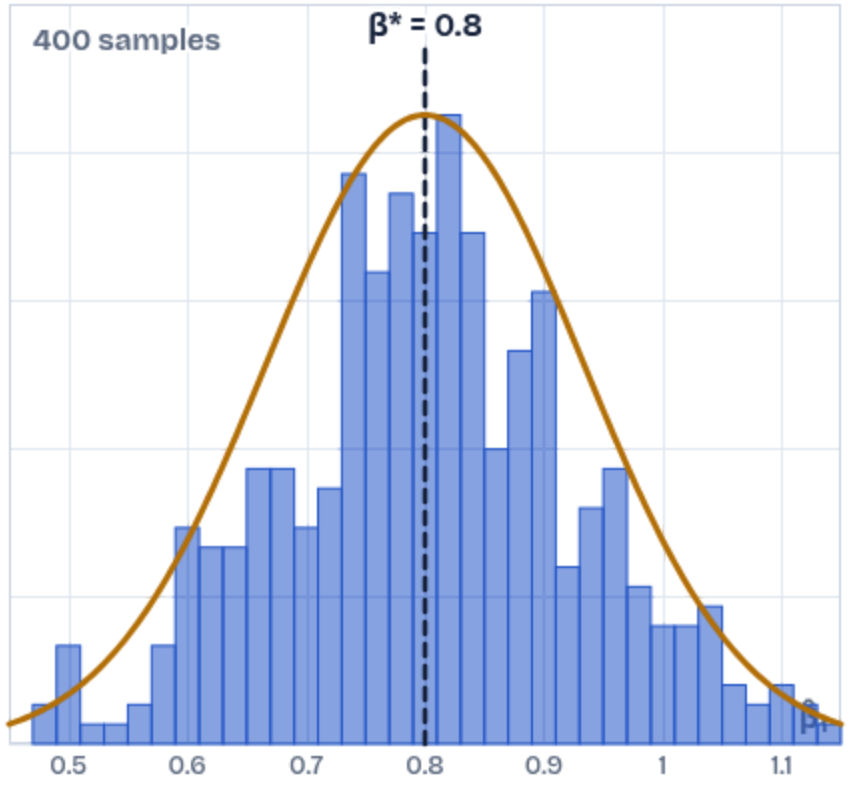
With $\mathbb{E}[\varepsilon \mid X] = 0$,

$$\boxed{\mathbb{E}[\hat{\beta} \mid X] = \beta}$$

With $\mathbf{Var}(\varepsilon \mid X) = \sigma^2 I_n$,

$$\boxed{\mathbf{Var}(\hat{\beta} \mid X) = \sigma^2 (X^\top X)^{-1}}$$

Diagonal entry j: $\mathbf{Var}(\hat{\beta}_j \mid X) = \sigma^2 \left[(X^\top X)^{-1}\right]_{jj}$



But we do not know σ^2

$$\mathbf{Var}(\hat{\beta}_j \mid X) = \sigma^2 \left[(X^\top X)^{-1} \right]_{jj}$$

contains the **unknown noise variance**.

Estimate it from the residuals: $\hat{\varepsilon}_i = y_i - \hat{y}_i$

Can we estimate σ^2 from the squared residuals?

Estimating the noise variance

$$\mathrm{RSS} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

$$\boxed{s^2 = \frac{\mathrm{RSS}}{n - p}}$$

p = number of fitted coefficients, *including the intercept*.

Why $n - p$? Fitting p coefficients uses p degrees of freedom.

Under the stated model, s^2 is unbiased for σ^2.

Standard error: definition

The **standard error** estimates the standard deviation of a coefficient estimator across repeated samples.

$$\boxed{SE(\hat{\beta}_j) = \sqrt{s^2 \left[(X^\top X)^{-1} \right]_{jj}}}$$

- Small SE ⇒ estimated more precisely
- Large SE ⇒ estimated less precisely

A small standard-error calculation

$$s^2 = 4, \qquad \left[(X^\top X)^{-1}\right]_{11} = 0.16$$

Exercise (1 minute): find $SE(\hat{\beta}_1)$.

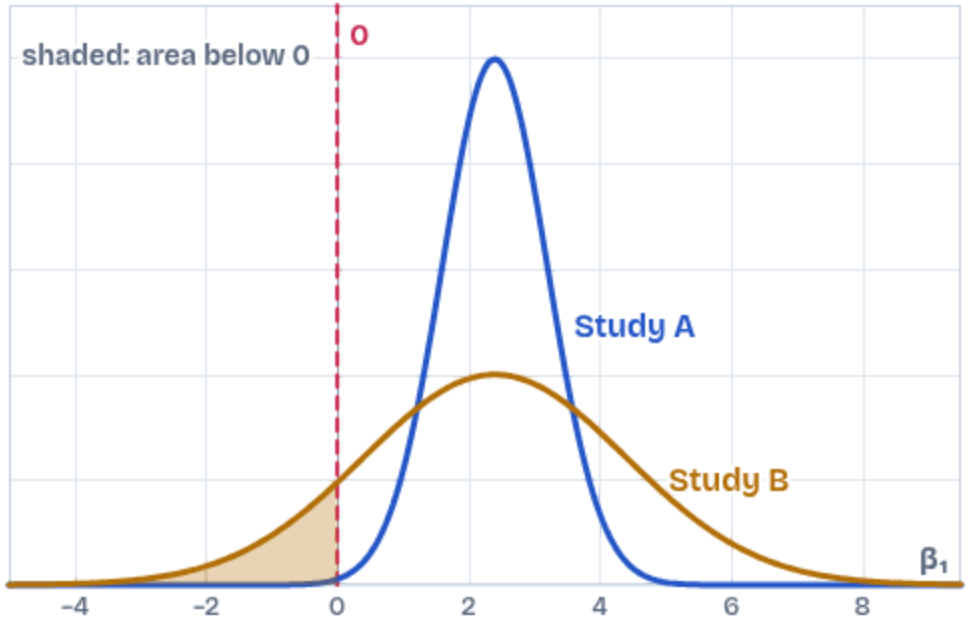
$$SE(\hat{\beta}_1) = \sqrt{4 \times 0.16} = \sqrt{0.64}$$

$$\boxed{SE(\hat{\beta}_1) = 0.8}$$

The SE measures uncertainty, not the coefficient's size.

Two estimates: which is more convincing?

	Study A	Study B
$\hat{\beta}_1$	2.4	2.4
$SE(\hat{\beta}_1)$	0.8	2.0



Which gives stronger evidence that the true coefficient differs from zero?

Study A: its estimate has a smaller standard error.

Hypothesis testing for a coefficient

Does feature x_j have a **nonzero partial regression coefficient**?

$$\boxed{\begin{aligned} H_0 &: \beta_j = 0 \\ H_1 &: \beta_j \neq 0 \end{aligned}}$$

Important: we test the true β_j, not whether $\hat{\beta}_j$ equals zero.

The t-statistic

To test $H_0 : \beta_j = b_0$,

$$t = \frac{\hat{\beta}_j - b_0}{SE(\hat{\beta}_j)}$$

For the common null $H_0 : \beta_j = 0$,

$$t = \frac{\hat{\beta}_j}{SE(\hat{\beta}_j)}$$

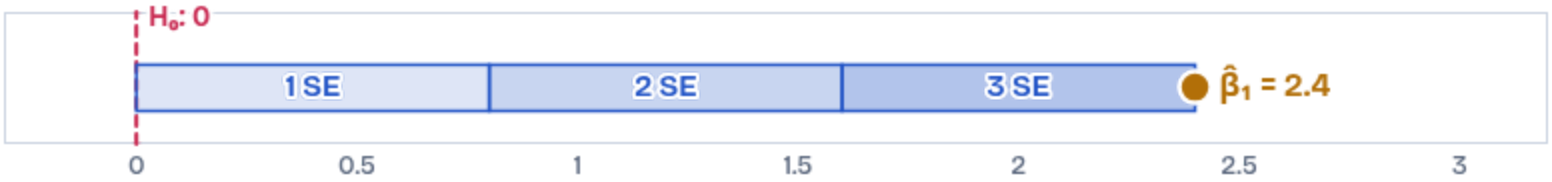
How many standard errors the estimate is from the null value

Exercise: compute the t-statistic

$$\hat{\beta}_1 = 2.4, \qquad SE(\hat{\beta}_1) = 0.8$$

Exercise (2 minutes): test $H_0 : \beta_1 = 0$.

$$t = \frac{2.4 - 0}{0.8} = \boxed{3}$$



The estimate is **3 standard errors from zero**.

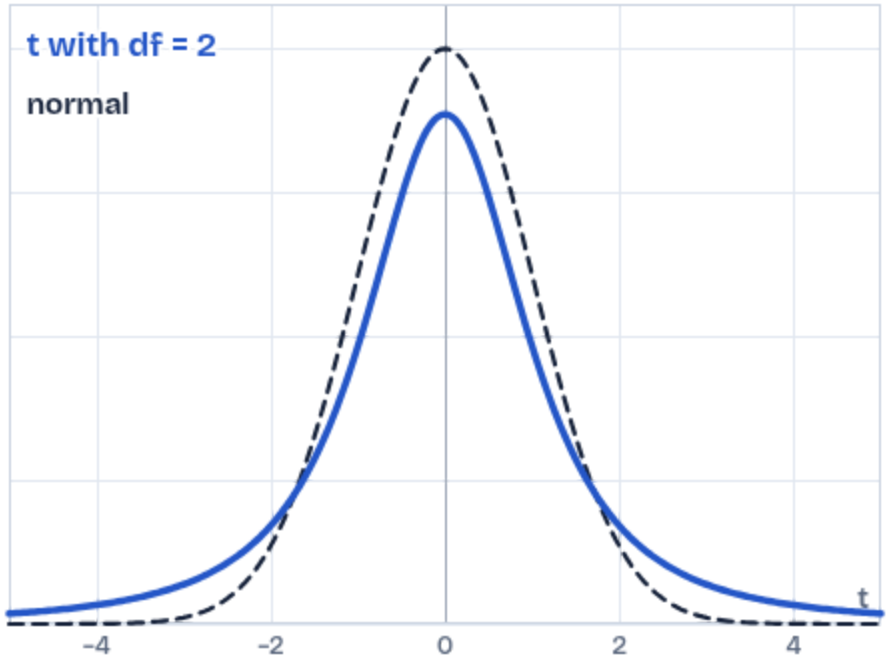
But is $t = 3$ large enough to reject H_0?

Why a t-distribution?

Classical model, **Gaussian errors**:

$$\frac{\hat{\beta}_j - \beta_j}{SE(\hat{\beta}_j)} \sim t_{n-p}$$

- n: observations
- p: fitted coefficients
- $n - p$: residual degrees of freedom



Heavier tails than the normal because σ^2 is estimated.

When do we reject the null hypothesis?

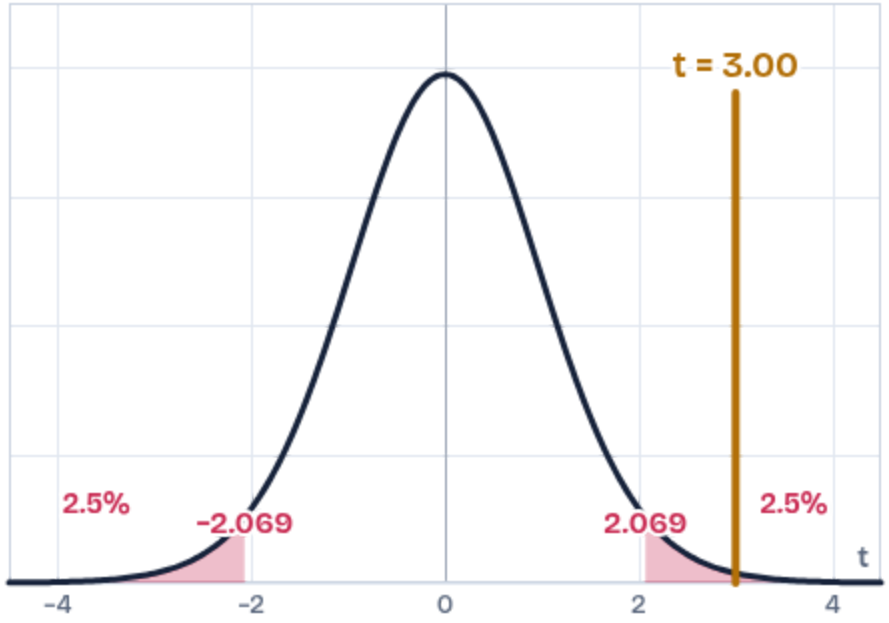
$n = 25,\ p = 2$:

$$\text{df} = n - p = 23$$

5% level, two-sided:

$$\pm\, t_{0.975,23} \approx \pm 2.069$$

$$\boxed{|3| > 2.069}$$

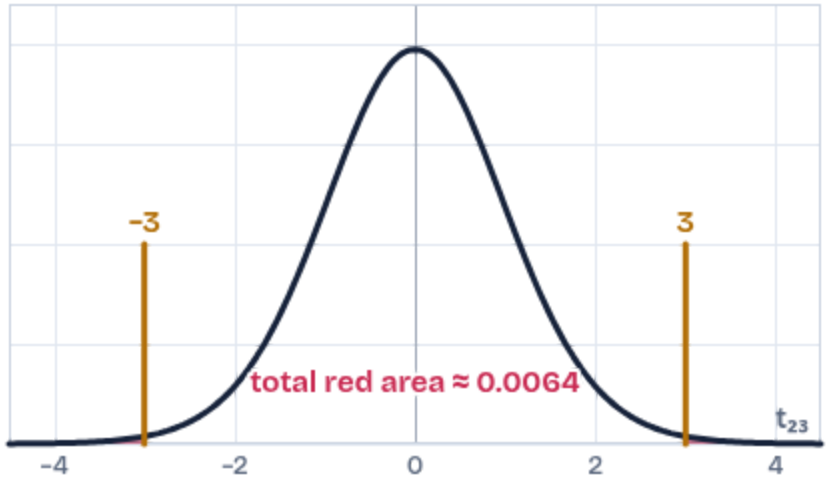


Reject $H_0 : \beta_1 = 0$ at the 5% level.

What is a p-value?

The probability, *assuming* H_0, of a test statistic at least as extreme as the one observed.

$$\boxed{p\text{-value} = P(|T_{23}| \geq 3) \approx 0.0064}$$



$0.0064 < 0.05$, so we reject H_0.

A p-value is not the probability that H_0 is true.

Confidence intervals: a different question

A t-test asks: "*Is the true coefficient plausibly zero?*"

A confidence interval asks: "*Which coefficient values are compatible with the data?*"

$$\hat{\beta}_j \pm t_{1-\alpha/2,\, n-p}\, SE(\hat{\beta}_j)$$

Exercise: construct a 95% confidence interval

$$n = 25,\ p = 2,\ t_{0.975,23} \approx 2.069, \quad \hat{\beta}_1 = 2.4,\ SE(\hat{\beta}_1) = 0.8$$

Exercise (2 minutes): find the 95% interval.

$$2.4 \pm 2.069 \times 0.8 = 2.4 \pm 1.6552$$

$$\boxed{CI_{95\%} \approx [0.745,\ 4.055]}$$

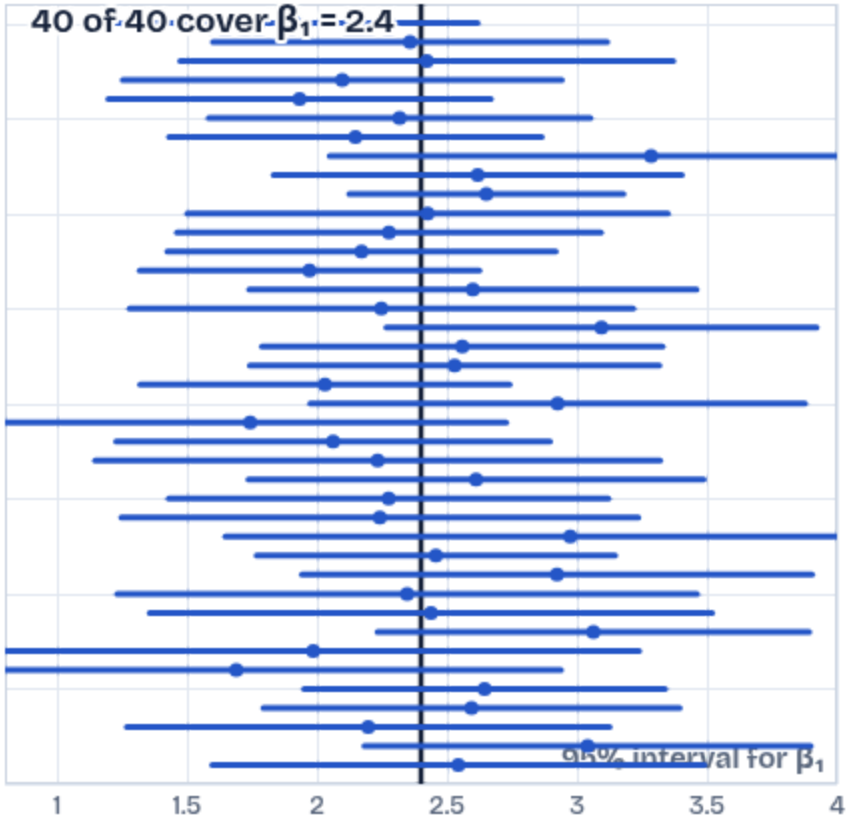


What does “95% confidence” mean?

Repeatedly collect new samples; build an interval from each.

About 95% of these intervals contain the fixed, true β_j.

It does **not** mean that, after seeing one interval, β_j has a 95% probability of lying inside it.



t-tests and confidence intervals agree

We tested $H_0 : \beta_1 = 0$ with estimate 2.4 and SE 0.8.

$$\boxed{CI_{95\%} = [0.745,\ 4.055]}$$

Does the interval contain 0?



No, so the two-sided t-test rejects H_0 at 5%.

Rejecting $H_0 \Leftrightarrow$ the null value lies outside the CI.

Connection to multicollinearity

$$\mathrm{Var}(\hat{\beta}_j \mid X) = \frac{\sigma^2}{\sum_i (x_{ij} - \bar{x}_j)^2} \underbrace{\frac{1}{1 - R_j^2}}_{\mathrm{VIF}_j}$$

Higher VIF ⇒ larger SE, *holding everything else fixed*.



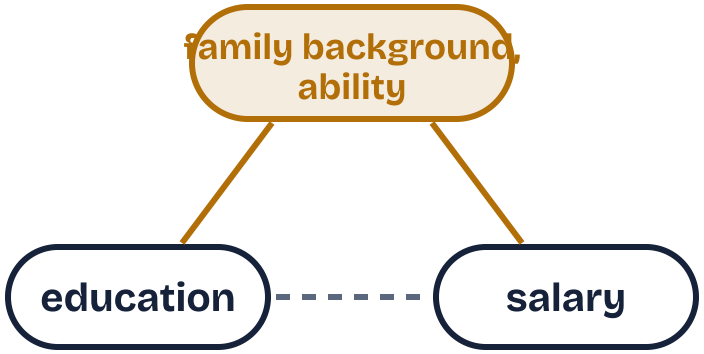
An imprecisely estimated coefficient may not be statistically significant.

Does statistical significance mean causation?

Education is significant in a regression predicting salary.

Have we proved that education causes higher salary?

No. Omitted variables and confounding can explain the association.



Significance ≠ causal identification ≠ practical importance.

Assumptions: when does OLS work?

For the familiar OLS formulas, assume:

1. $\mathbb{E}[\varepsilon \mid X] = 0$: zero conditional mean
2. $\mathbf{rank}(X) = p$: no perfect multicollinearity

Then OLS is unbiased conditional on X.

A high R^2 does not verify these assumptions.

Additional assumptions for classical inference

For the conventional variance formula: $\mathbf{Var}(\varepsilon \mid X) = \sigma^2 I_n$

- **Homoscedasticity:** constant conditional error variance
- **Uncorrelated errors:** zero conditional covariance across observations

For exact small-sample t-tests and intervals, also:

$$\varepsilon \mid X \sim \mathcal{N}(0, \sigma^2 I_n)$$

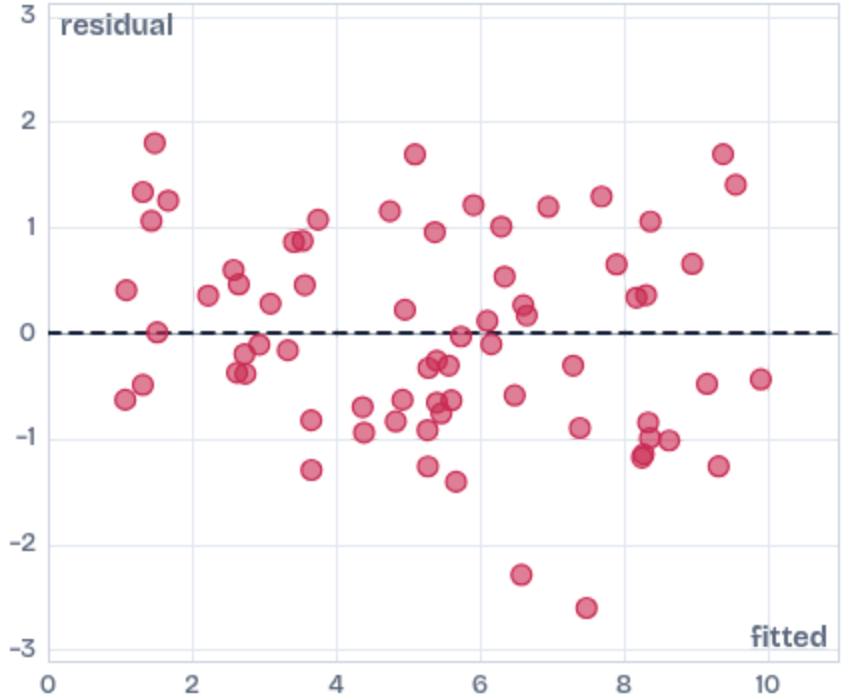
Residual diagnostics

$$\boxed{e_i = y_i - \hat{y}_i}$$

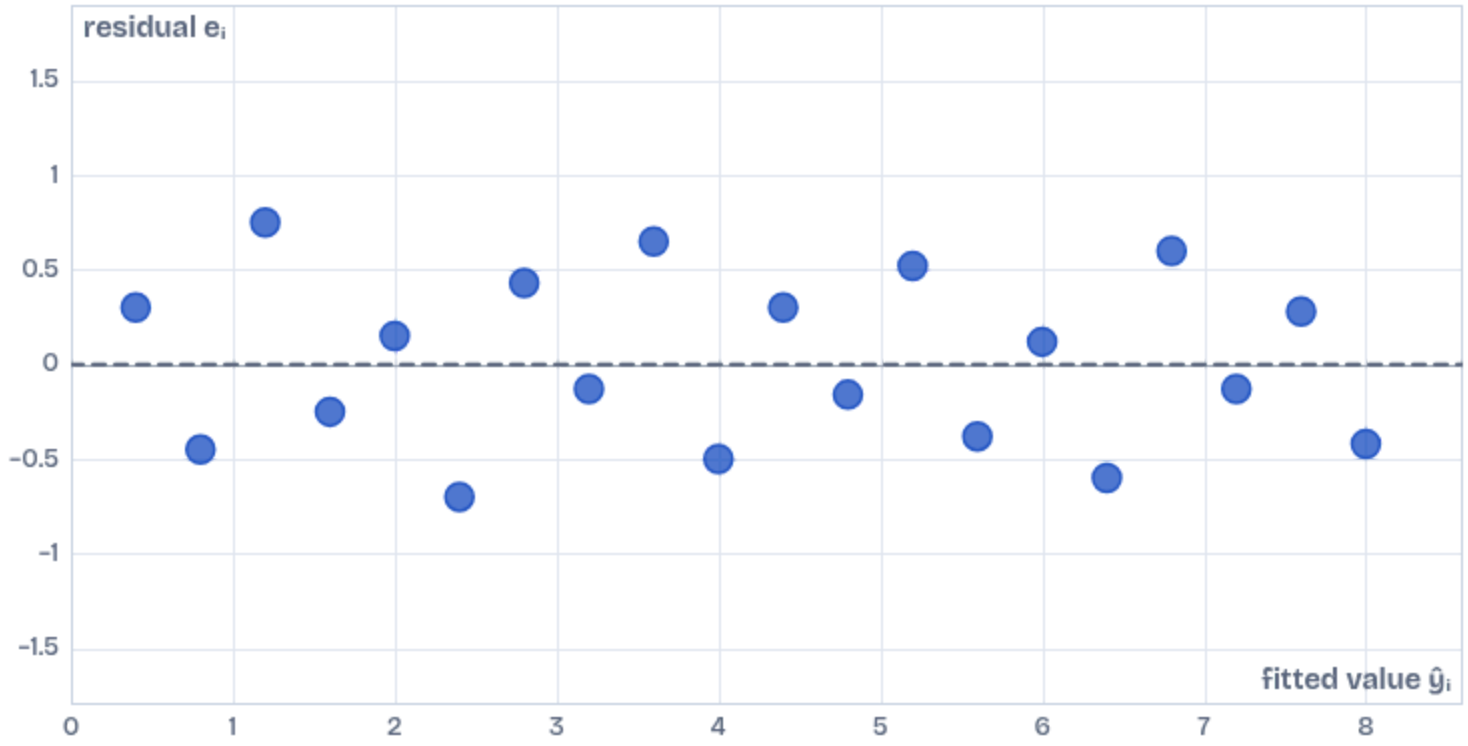
Plot e_i against fitted $\hat{y}_i$.

- Random scatter near zero is reassuring.
- Systematic shapes reveal problems.

What patterns should we look for?

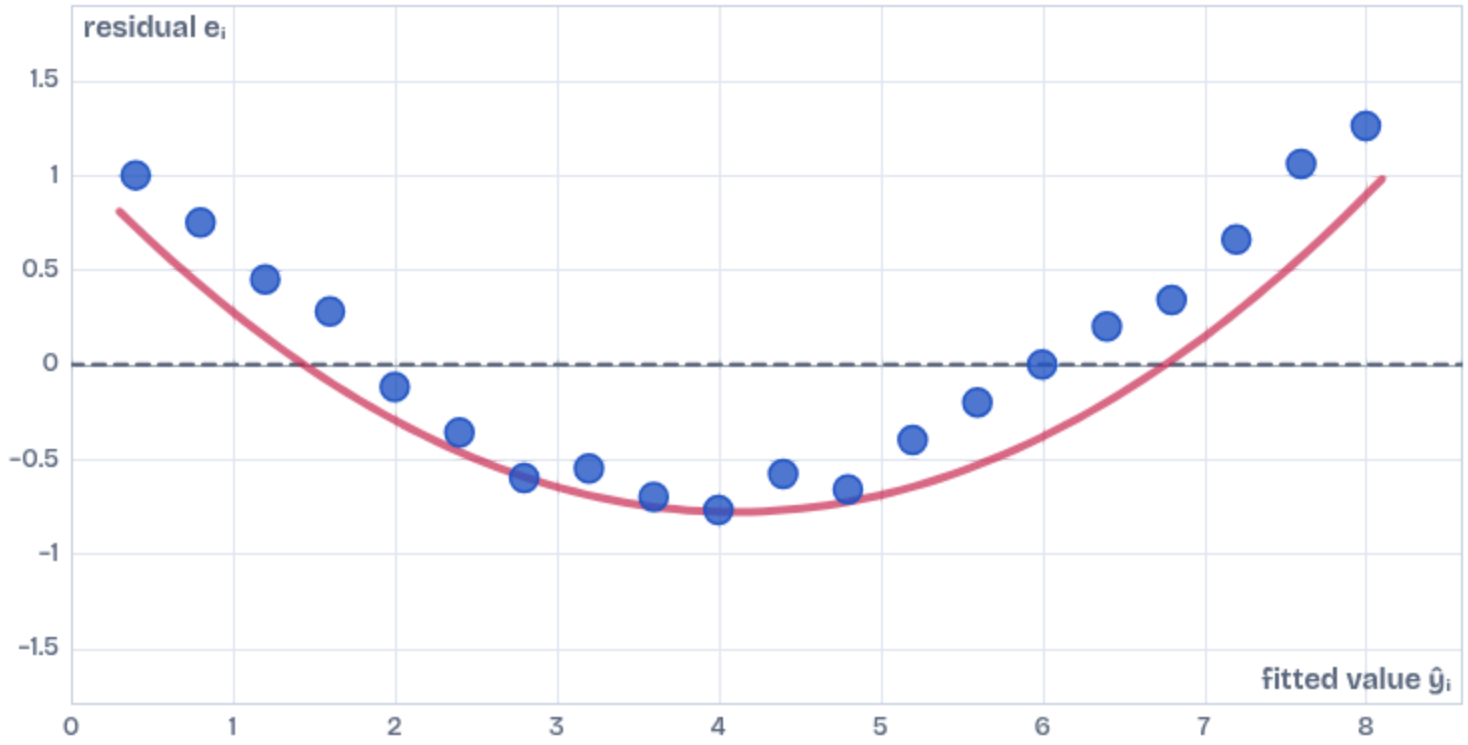


Residual plot A: no clear pattern



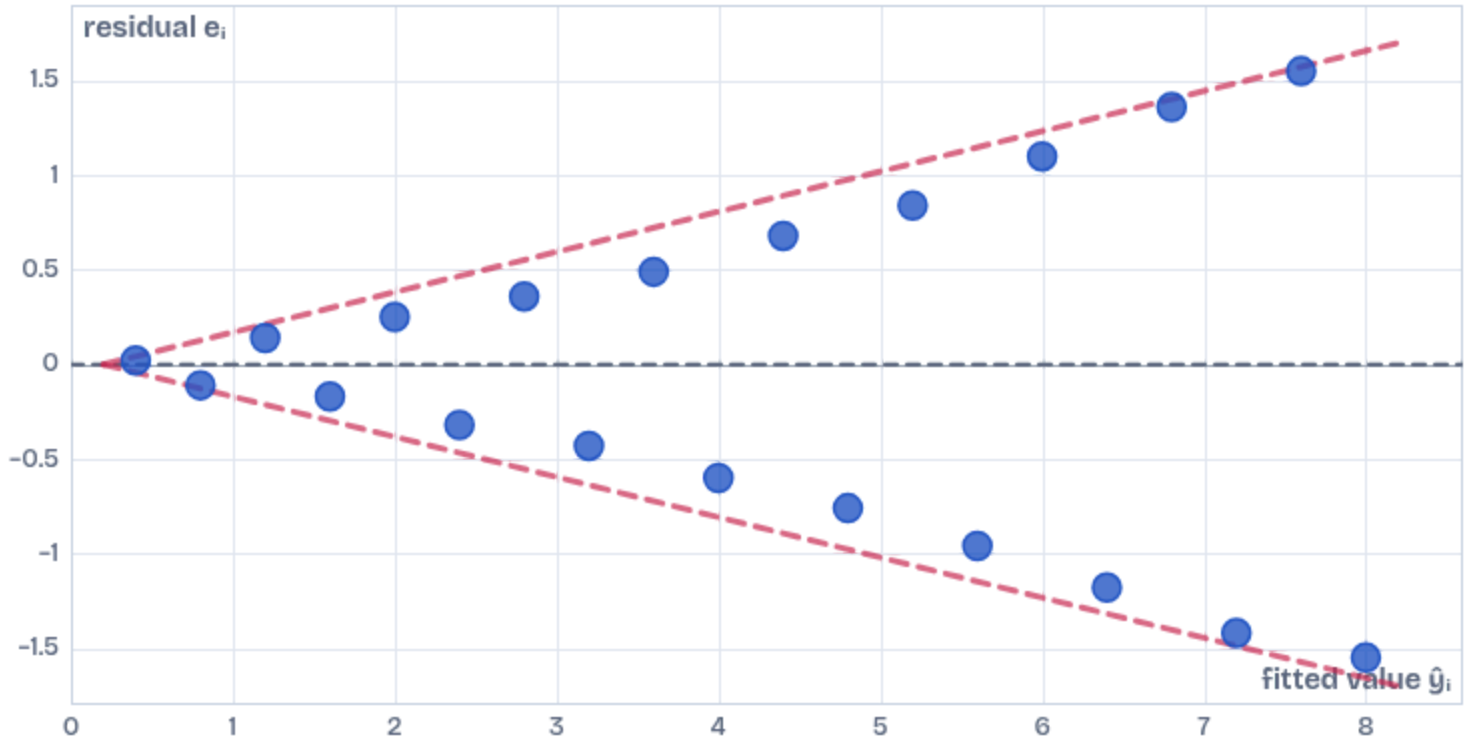
No obvious curvature or change in spread.

Residual plot B: curved pattern



Possible misspecification of the conditional mean (e.g. missing nonlinearity).

Residual plot C: funnel shape



Possible heteroscedasticity: error variance changes with the fitted value.

Can residual plots prove the model is correct?

No. Residual plots are **diagnostic tools**, not proofs.

- Curvature ⇒ possible mean-model misspecification
- Funnel ⇒ possible unequal error variance

Other checks are needed for dependence and error normality.

A visually random residual plot does not establish causality.

Final check: what does each quantity tell us?

Quantity	Meaning
$\hat{\beta}_j$	Estimated association (given the other features)
$SE(\hat{\beta}_j)$	Precision of the estimate
t-statistic	Distance from the null, in standard errors
95% CI	Range from a 95% coverage procedure
Residual plot	Check for visible misspecification

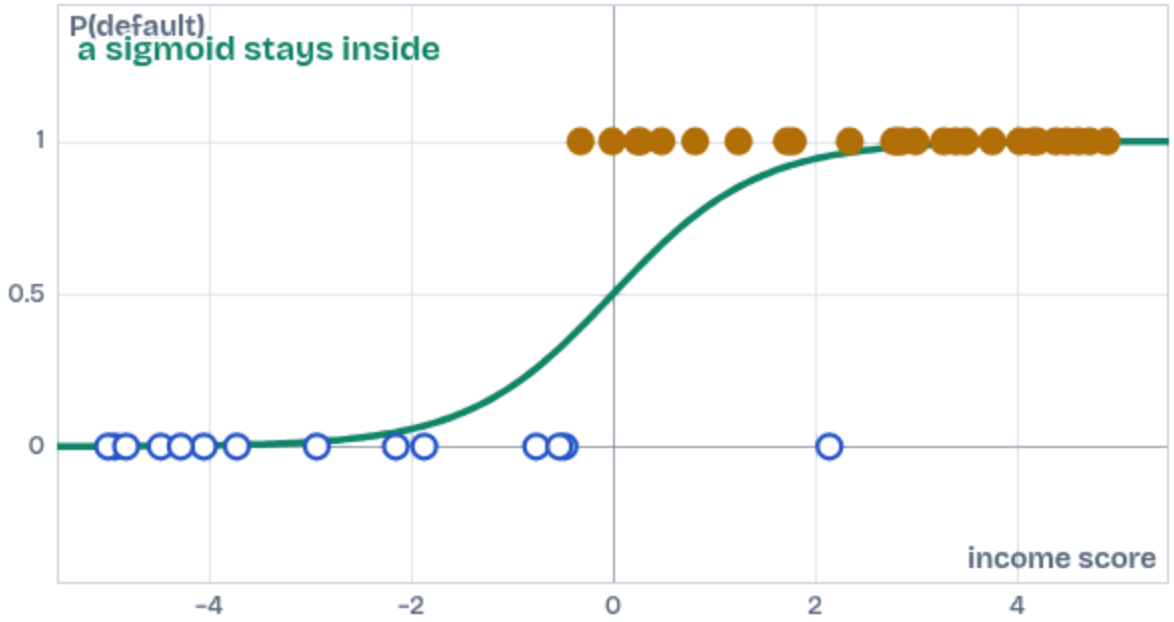
A useful model needs both predictive assessment and trustworthy inference.

Takeaways

1. OLS is an orthogonal projection.
2. Gaussian MLE gives the same coefficients.
3. Good predictions do not guarantee identifiable coefficients.
4. Reliable inference requires assumptions and diagnostics.

Next lecture

What if the target is a class label, not a real number?



Logistic regression